pcbai assembly reference

DC0_v2

2 copper layers | 100 x 100 mm | 145 fitted items | 0 DNP sites

One complete working stack is DC0 + DC1a/b/c/d + DC2 + DC3 + motherboard. Quantities in this reference are for that stack, not all spare bare PCBs in the JLC batches. All assembly is by hand.

Hardware on hand / ordered: mostly 5% 0805 resistors, TL074 SOIC-14, 74HCT595 SOP-16 (HCT, not HC), 100nF 0805 capacitors, and 1N4148W SOD-123 diodes. Board/CSV values below are preserved exactly, not rounded to purchased kit values. CSV tolerance requirements are shown where provided.

DC0 board labels its eight shift registers 74AHCT595S16-13. The ordered HCT part is noted here but the BOM does not silently rename it. 0.1uF = 100nF. Straddle connectors are counted as connector items, not bare copper contacts.

Verified source: /home/kit/dc0-pads-v2/DC0_verified.kicad_pcb
SHA-256 899da15d132ea29582af96c94f942296de6b0a254dbfdaa3360ae32ee56a69ea
Mounting holes (excluded from parts totals): H2 at (95, 5) mm; H1 at (5, 5) mm.

Grouped fitting summary

Value	Package	Qty	CSV tolerance
0.1uF	0805 capacitor	8	-
1uF	0805 capacitor	64	-
10k	0805 resistor	64	-
2x38_2.54mm_GENERIC_MANUAL_STRADDLE	2x38 straddle, 2.54 mm	1	-
74AHCT595S16-13	SOIC-16 / SOP-16, 1.27 mm	8	-

Every populated component

Ref	Value	Footprint / package	Side	Tol / status
C1	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C2	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C3	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C4	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C5	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C6	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C7	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C8	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
CP0	1uF	C_0805_2012Metric	F	FIT
CP1	1uF	C_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
CP2	1uF	C_0805_2012Metric	F	FIT
CP3	1uF	C_0805_2012Metric	F	FIT
CP4	1uF	C_0805_2012Metric	F	FIT
CP5	1uF	C_0805_2012Metric	F	FIT
CP6	1uF	C_0805_2012Metric	F	FIT
CP7	1uF	C_0805_2012Metric	F	FIT
CP8	1uF	C_0805_2012Metric	F	FIT
CP9	1uF	C_0805_2012Metric	F	FIT
CP10	1uF	C_0805_2012Metric	F	FIT
CP11	1uF	C_0805_2012Metric	F	FIT
CP12	1uF	C_0805_2012Metric	F	FIT
CP13	1uF	C_0805_2012Metric	F	FIT
CP14	1uF	C_0805_2012Metric	F	FIT
CP15	1uF	C_0805_2012Metric	F	FIT
CP16	1uF	C_0805_2012Metric	F	FIT
CP17	1uF	C_0805_2012Metric	F	FIT
CP18	1uF	C_0805_2012Metric	F	FIT
CP19	1uF	C_0805_2012Metric	F	FIT
CP20	1uF	C_0805_2012Metric	F	FIT
CP21	1uF	C_0805_2012Metric	F	FIT
CP22	1uF	C_0805_2012Metric	F	FIT
CP23	1uF	C_0805_2012Metric	F	FIT
CP24	1uF	C_0805_2012Metric	F	FIT
CP25	1uF	C_0805_2012Metric	F	FIT
CP26	1uF	C_0805_2012Metric	F	FIT
CP27	1uF	C_0805_2012Metric	F	FIT
CP28	1uF	C_0805_2012Metric	F	FIT
CP29	1uF	C_0805_2012Metric	F	FIT
CP30	1uF	C_0805_2012Metric	F	FIT
CP31	1uF	C_0805_2012Metric	F	FIT
CP32	1uF	C_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
CP33	1uF	C_0805_2012Metric	F	FIT
CP34	1uF	C_0805_2012Metric	F	FIT
CP35	1uF	C_0805_2012Metric	F	FIT
CP36	1uF	C_0805_2012Metric	F	FIT
CP37	1uF	C_0805_2012Metric	F	FIT
CP38	1uF	C_0805_2012Metric	F	FIT
CP39	1uF	C_0805_2012Metric	F	FIT
CP40	1uF	C_0805_2012Metric	F	FIT
CP41	1uF	C_0805_2012Metric	F	FIT
CP42	1uF	C_0805_2012Metric	F	FIT
CP43	1uF	C_0805_2012Metric	F	FIT
CP44	1uF	C_0805_2012Metric	F	FIT
CP45	1uF	C_0805_2012Metric	F	FIT
CP46	1uF	C_0805_2012Metric	F	FIT
CP47	1uF	C_0805_2012Metric	F	FIT
CP48	1uF	C_0805_2012Metric	F	FIT
CP49	1uF	C_0805_2012Metric	F	FIT
CP50	1uF	C_0805_2012Metric	F	FIT
CP51	1uF	C_0805_2012Metric	F	FIT
CP52	1uF	C_0805_2012Metric	F	FIT
CP53	1uF	C_0805_2012Metric	F	FIT
CP54	1uF	C_0805_2012Metric	F	FIT
CP55	1uF	C_0805_2012Metric	F	FIT
CP56	1uF	C_0805_2012Metric	F	FIT
CP57	1uF	C_0805_2012Metric	F	FIT
CP58	1uF	C_0805_2012Metric	F	FIT
CP59	1uF	C_0805_2012Metric	F	FIT
CP60	1uF	C_0805_2012Metric	F	FIT
CP61	1uF	C_0805_2012Metric	F	FIT
CP62	1uF	C_0805_2012Metric	F	FIT
CP63	1uF	C_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
J0	2x38_2.54mm_GENERIC_MANUAL_STRADDLE	DC0_Isolated:Straddle2x38	F	FIT
RP0	10k	R_0805_2012Metric	F	FIT
RP1	10k	R_0805_2012Metric	F	FIT
RP2	10k	R_0805_2012Metric	F	FIT
RP3	10k	R_0805_2012Metric	F	FIT
RP4	10k	R_0805_2012Metric	F	FIT
RP5	10k	R_0805_2012Metric	F	FIT
RP6	10k	R_0805_2012Metric	F	FIT
RP7	10k	R_0805_2012Metric	F	FIT
RP8	10k	R_0805_2012Metric	F	FIT
RP9	10k	R_0805_2012Metric	F	FIT
RP10	10k	R_0805_2012Metric	F	FIT
RP11	10k	R_0805_2012Metric	F	FIT
RP12	10k	R_0805_2012Metric	F	FIT
RP13	10k	R_0805_2012Metric	F	FIT
RP14	10k	R_0805_2012Metric	F	FIT
RP15	10k	R_0805_2012Metric	F	FIT
RP16	10k	R_0805_2012Metric	F	FIT
RP17	10k	R_0805_2012Metric	F	FIT
RP18	10k	R_0805_2012Metric	F	FIT
RP19	10k	R_0805_2012Metric	F	FIT
RP20	10k	R_0805_2012Metric	F	FIT
RP21	10k	R_0805_2012Metric	F	FIT
RP22	10k	R_0805_2012Metric	F	FIT
RP23	10k	R_0805_2012Metric	F	FIT
RP24	10k	R_0805_2012Metric	F	FIT
RP25	10k	R_0805_2012Metric	F	FIT
RP26	10k	R_0805_2012Metric	F	FIT
RP27	10k	R_0805_2012Metric	F	FIT
RP28	10k	R_0805_2012Metric	F	FIT
RP29	10k	R_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
RP30	10k	R_0805_2012Metric	F	FIT
RP31	10k	R_0805_2012Metric	F	FIT
RP32	10k	R_0805_2012Metric	F	FIT
RP33	10k	R_0805_2012Metric	F	FIT
RP34	10k	R_0805_2012Metric	F	FIT
RP35	10k	R_0805_2012Metric	F	FIT
RP36	10k	R_0805_2012Metric	F	FIT
RP37	10k	R_0805_2012Metric	F	FIT
RP38	10k	R_0805_2012Metric	F	FIT
RP39	10k	R_0805_2012Metric	F	FIT
RP40	10k	R_0805_2012Metric	F	FIT
RP41	10k	R_0805_2012Metric	F	FIT
RP42	10k	R_0805_2012Metric	F	FIT
RP43	10k	R_0805_2012Metric	F	FIT
RP44	10k	R_0805_2012Metric	F	FIT
RP45	10k	R_0805_2012Metric	F	FIT
RP46	10k	R_0805_2012Metric	F	FIT
RP47	10k	R_0805_2012Metric	F	FIT
RP48	10k	R_0805_2012Metric	F	FIT
RP49	10k	R_0805_2012Metric	F	FIT
RP50	10k	R_0805_2012Metric	F	FIT
RP51	10k	R_0805_2012Metric	F	FIT
RP52	10k	R_0805_2012Metric	F	FIT
RP53	10k	R_0805_2012Metric	F	FIT
RP54	10k	R_0805_2012Metric	F	FIT
RP55	10k	R_0805_2012Metric	F	FIT
RP56	10k	R_0805_2012Metric	F	FIT
RP57	10k	R_0805_2012Metric	F	FIT
RP58	10k	R_0805_2012Metric	F	FIT
RP59	10k	R_0805_2012Metric	F	FIT
RP60	10k	R_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
RP61	10k	R_0805_2012Metric	F	FIT
RP62	10k	R_0805_2012Metric	F	FIT
RP63	10k	R_0805_2012Metric	F	FIT
U1	74AHCT595S16-13	Package_SO:SOIC-16_3.9x9.9mm_P1.27mm	F	FIT
U2	74AHCT595S16-13	Package_SO:SOIC-16_3.9x9.9mm_P1.27mm	F	FIT
U3	74AHCT595S16-13	Package_SO:SOIC-16_3.9x9.9mm_P1.27mm	F	FIT
U4	74AHCT595S16-13	Package_SO:SOIC-16_3.9x9.9mm_P1.27mm	F	FIT
U5	74AHCT595S16-13	Package_SO:SOIC-16_3.9x9.9mm_P1.27mm	F	FIT
U6	74AHCT595S16-13	Package_SO:SOIC-16_3.9x9.9mm_P1.27mm	F	FIT
U7	74AHCT595S16-13	Package_SO:SOIC-16_3.9x9.9mm_P1.27mm	F	FIT
U8	74AHCT595S16-13	Package_SO:SOIC-16_3.9x9.9mm_P1.27mm	F	FIT

DC1a

4 copper layers | 100 x 100 mm | 195 fitted items | 341 DNP sites

Populate ONLY variant DC1a for neurons N0 through N3. Same bare 4-layer board for all four variants. CSV overrides universal footprint values at all 532 resistor sites. There are 112 extra sites DNP on every variant; total DNP is larger. DNP means leave open, never fit 0 ohms.

Verified source: /home/kit/dc1-universal/fix12/board.kicad_pcb
SHA-256 b5d0b202d140751f72bbe6a50ee582ed0e2a69545f96704e8a4b415b7e787748
Population: /home/kit/dc1-universal/fix12/DC1a.csv | SHA-256 ba47456b6f6b00546de66d44b820a96686a5c440bf5007a556305e7b4a61b7f8
Mounting holes (excluded from parts totals): H1 at (5, 5) mm; H2 at (95, 5) mm.

Grouped fitting summary

Value	Package	Qty	CSV tolerance
0.1uF	0805 capacitor	2	-
1.1M	0805 resistor	4	5.0%
1.2M	0805 resistor	3	5.0%
1.3M	0805 resistor	1	5.0%
1.5M	0805 resistor	1	5.0%
1.6M	0805 resistor	1	5.0%
100k	0805 resistor	4	1.0%
100k	0805 resistor	3	5.0%
110k	0805 resistor	9	5.0%
120k	0805 resistor	3	5.0%
130k	0805 resistor	6	5.0%
14.7k	0805 resistor	1	1.0%
150k	0805 resistor	8	5.0%
160k	0805 resistor	6	5.0%
18.2k	0805 resistor	1	1.0%
180k	0805 resistor	7	5.0%
1M	0805 resistor	2	5.0%
2.2M	0805 resistor	1	5.0%
2.4M	0805 resistor	1	5.0%
200k	0805 resistor	9	5.0%
22.6k	0805 resistor	1	1.0%
220k	0805 resistor	7	5.0%
240k	0805 resistor	8	5.0%

Value	Package	Qty	CSV tolerance
270k	0805 resistor	6	5.0%
27k	0805 resistor	1	5.0%
29.4k	0805 resistor	1	1.0%
2M	0805 resistor	3	5.0%
3.3M	0805 resistor	2	5.0%
300k	0805 resistor	4	5.0%
30k	0805 resistor	1	5.0%
330k	0805 resistor	12	5.0%
360k	0805 resistor	7	5.0%
390k	0805 resistor	9	5.0%
430k	0805 resistor	7	5.0%
43k	0805 resistor	1	5.0%
470k	0805 resistor	7	5.0%
47k	0805 resistor	3	5.0%
510k	0805 resistor	2	5.0%
560k	0805 resistor	6	5.0%
56k	0805 resistor	2	5.0%
6.2M	0805 resistor	1	5.0%
620k	0805 resistor	3	5.0%
62k	0805 resistor	1	5.0%
680k	0805 resistor	6	5.0%
68k	0805 resistor	1	5.0%
750k	0805 resistor	1	5.0%
75k	0805 resistor	5	5.0%
82k	0805 resistor	3	5.0%
910k	0805 resistor	3	5.0%
91k	0805 resistor	6	5.0%
PinHeader_2x38_Straddle	2x38 straddle, 2.54 mm	1	-
TL074	SOIC-14, 1.27 mm	1	-

Every populated component

Ref	Value	Footprint / package	Side	Tol / status
C_U1N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U1P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
J_STRADDLE_EDGE	PinHeader_2x38_Straddle	Connector_EdgeMount:PinHeader_2x38_Straddle_P2.54mm	F	FIT
R002	30k	R_0805_2012Metric	F	5.0%
R008	47k	R_0805_2012Metric	F	5.0%
R009	390k	R_0805_2012Metric	F	5.0%
R012	110k	R_0805_2012Metric	F	5.0%
R013	2M	R_0805_2012Metric	F	5.0%
R015	160k	R_0805_2012Metric	F	5.0%
R018	1.5M	R_0805_2012Metric	F	5.0%
R020	120k	R_0805_2012Metric	F	5.0%
R021	180k	R_0805_2012Metric	F	5.0%
R026	680k	R_0805_2012Metric	F	5.0%
R027	160k	R_0805_2012Metric	F	5.0%
R029	130k	R_0805_2012Metric	F	5.0%
R032	510k	R_0805_2012Metric	F	5.0%
R033	150k	R_0805_2012Metric	F	5.0%
R035	360k	R_0805_2012Metric	F	5.0%
R038	110k	R_0805_2012Metric	F	5.0%
R040	240k	R_0805_2012Metric	F	5.0%
R042	270k	R_0805_2012Metric	F	5.0%
R044	200k	R_0805_2012Metric	F	5.0%
R046	430k	R_0805_2012Metric	F	5.0%
R050	150k	R_0805_2012Metric	F	5.0%
R052	180k	R_0805_2012Metric	F	5.0%
R053	360k	R_0805_2012Metric	F	5.0%
R056	82k	R_0805_2012Metric	F	5.0%
R059	130k	R_0805_2012Metric	F	5.0%
R060	200k	R_0805_2012Metric	F	5.0%
R062	360k	R_0805_2012Metric	F	5.0%
R064	240k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R066	470k	R_0805_2012Metric	F	5.0%
R067	390k	R_0805_2012Metric	F	5.0%
R070	560k	R_0805_2012Metric	F	5.0%
R071	620k	R_0805_2012Metric	F	5.0%
R076	680k	R_0805_2012Metric	F	5.0%
R079	160k	R_0805_2012Metric	F	5.0%
R084	220k	R_0805_2012Metric	F	5.0%
R086	360k	R_0805_2012Metric	F	5.0%
R087	3.3M	R_0805_2012Metric	F	5.0%
R090	270k	R_0805_2012Metric	F	5.0%
R093	330k	R_0805_2012Metric	F	5.0%
R094	1.1M	R_0805_2012Metric	F	5.0%
R095	150k	R_0805_2012Metric	F	5.0%
R098	300k	R_0805_2012Metric	F	5.0%
R104	160k	R_0805_2012Metric	F	5.0%
R106	47k	R_0805_2012Metric	F	5.0%
R112	220k	R_0805_2012Metric	F	5.0%
R114	56k	R_0805_2012Metric	F	5.0%
R116	910k	R_0805_2012Metric	F	5.0%
R119	160k	R_0805_2012Metric	F	5.0%
R121	91k	R_0805_2012Metric	F	5.0%
R122	82k	R_0805_2012Metric	F	5.0%
R126	220k	R_0805_2012Metric	F	5.0%
R127	130k	R_0805_2012Metric	F	5.0%
R128	430k	R_0805_2012Metric	F	5.0%
R131	180k	R_0805_2012Metric	F	5.0%
R132	2.2M	R_0805_2012Metric	F	5.0%
R133	120k	R_0805_2012Metric	F	5.0%
R136	100k	R_0805_2012Metric	F	5.0%
R137	430k	R_0805_2012Metric	F	5.0%
R138	130k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R141	91k	R_0805_2012Metric	F	5.0%
R142	2M	R_0805_2012Metric	F	5.0%
R144	180k	R_0805_2012Metric	F	5.0%
R147	47k	R_0805_2012Metric	F	5.0%
R148	560k	R_0805_2012Metric	F	5.0%
R149	150k	R_0805_2012Metric	F	5.0%
R153	240k	R_0805_2012Metric	F	5.0%
R154	75k	R_0805_2012Metric	F	5.0%
R156	430k	R_0805_2012Metric	F	5.0%
R162	75k	R_0805_2012Metric	F	5.0%
R164	390k	R_0805_2012Metric	F	5.0%
R167	270k	R_0805_2012Metric	F	5.0%
R168	150k	R_0805_2012Metric	F	5.0%
R170	160k	R_0805_2012Metric	F	5.0%
R173	330k	R_0805_2012Metric	F	5.0%
R174	150k	R_0805_2012Metric	F	5.0%
R177	300k	R_0805_2012Metric	F	5.0%
R180	680k	R_0805_2012Metric	F	5.0%
R181	620k	R_0805_2012Metric	F	5.0%
R183	240k	R_0805_2012Metric	F	5.0%
R186	200k	R_0805_2012Metric	F	5.0%
R187	330k	R_0805_2012Metric	F	5.0%
R188	1.6M	R_0805_2012Metric	F	5.0%
R192	330k	R_0805_2012Metric	F	5.0%
R193	560k	R_0805_2012Metric	F	5.0%
R194	620k	R_0805_2012Metric	F	5.0%
R197	75k	R_0805_2012Metric	F	5.0%
R198	220k	R_0805_2012Metric	F	5.0%
R199	180k	R_0805_2012Metric	F	5.0%
R200	360k	R_0805_2012Metric	F	5.0%
R202	91k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R203	910k	R_0805_2012Metric	F	5.0%
R205	27k	R_0805_2012Metric	F	5.0%
R207	1M	R_0805_2012Metric	F	5.0%
R209	110k	R_0805_2012Metric	F	5.0%
R210	180k	R_0805_2012Metric	F	5.0%
R212	91k	R_0805_2012Metric	F	5.0%
R214	1.3M	R_0805_2012Metric	F	5.0%
R216	180k	R_0805_2012Metric	F	5.0%
R218	390k	R_0805_2012Metric	F	5.0%
R221	680k	R_0805_2012Metric	F	5.0%
R223	200k	R_0805_2012Metric	F	5.0%
R224	220k	R_0805_2012Metric	F	5.0%
R226	330k	R_0805_2012Metric	F	5.0%
R228	470k	R_0805_2012Metric	F	5.0%
R230	270k	R_0805_2012Metric	F	5.0%
R235	1.2M	R_0805_2012Metric	F	5.0%
R237	470k	R_0805_2012Metric	F	5.0%
R239	360k	R_0805_2012Metric	F	5.0%
R241	330k	R_0805_2012Metric	F	5.0%
R245	430k	R_0805_2012Metric	F	5.0%
R246	300k	R_0805_2012Metric	F	5.0%
R247	110k	R_0805_2012Metric	F	5.0%
R252	75k	R_0805_2012Metric	F	5.0%
R254	150k	R_0805_2012Metric	F	5.0%
R256	510k	R_0805_2012Metric	F	5.0%
R260	43k	R_0805_2012Metric	F	5.0%
R261	330k	R_0805_2012Metric	F	5.0%
R263	110k	R_0805_2012Metric	F	5.0%
R265	240k	R_0805_2012Metric	F	5.0%
R266	390k	R_0805_2012Metric	F	5.0%
R269	150k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R271	390k	R_0805_2012Metric	F	5.0%
R274	91k	R_0805_2012Metric	F	5.0%
R276	750k	R_0805_2012Metric	F	5.0%
R277	1.2M	R_0805_2012Metric	F	5.0%
R280	330k	R_0805_2012Metric	F	5.0%
R282	2.4M	R_0805_2012Metric	F	5.0%
R285	390k	R_0805_2012Metric	F	5.0%
R289	2M	R_0805_2012Metric	F	5.0%
R293	300k	R_0805_2012Metric	F	5.0%
R295	1M	R_0805_2012Metric	F	5.0%
R296	68k	R_0805_2012Metric	F	5.0%
R301	110k	R_0805_2012Metric	F	5.0%
R303	110k	R_0805_2012Metric	F	5.0%
R308	240k	R_0805_2012Metric	F	5.0%
R309	470k	R_0805_2012Metric	F	5.0%
R310	240k	R_0805_2012Metric	F	5.0%
R312	110k	R_0805_2012Metric	F	5.0%
R313	270k	R_0805_2012Metric	F	5.0%
R314	470k	R_0805_2012Metric	F	5.0%
R315	330k	R_0805_2012Metric	F	5.0%
R318	680k	R_0805_2012Metric	F	5.0%
R319	910k	R_0805_2012Metric	F	5.0%
R320	560k	R_0805_2012Metric	F	5.0%
R323	680k	R_0805_2012Metric	F	5.0%
R325	330k	R_0805_2012Metric	F	5.0%
R326	470k	R_0805_2012Metric	F	5.0%
R328	560k	R_0805_2012Metric	F	5.0%
R331	1.2M	R_0805_2012Metric	F	5.0%
R332	3.3M	R_0805_2012Metric	F	5.0%
R335	560k	R_0805_2012Metric	F	5.0%
R337	200k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R339	390k	R_0805_2012Metric	F	5.0%
R341	1.1M	R_0805_2012Metric	F	5.0%
R342	430k	R_0805_2012Metric	F	5.0%
R344	270k	R_0805_2012Metric	F	5.0%
R346	430k	R_0805_2012Metric	F	5.0%
R348	82k	R_0805_2012Metric	F	5.0%
R352	200k	R_0805_2012Metric	F	5.0%
R354	91k	R_0805_2012Metric	F	5.0%
R355	62k	R_0805_2012Metric	F	5.0%
R360	75k	R_0805_2012Metric	F	5.0%
R365	220k	R_0805_2012Metric	F	5.0%
R367	470k	R_0805_2012Metric	F	5.0%
R371	390k	R_0805_2012Metric	F	5.0%
R374	360k	R_0805_2012Metric	F	5.0%
R376	200k	R_0805_2012Metric	F	5.0%
R377	200k	R_0805_2012Metric	F	5.0%
R379	1.1M	R_0805_2012Metric	F	5.0%
R381	130k	R_0805_2012Metric	F	5.0%
R383	220k	R_0805_2012Metric	F	5.0%
R384	330k	R_0805_2012Metric	F	5.0%
R386	240k	R_0805_2012Metric	F	5.0%
R390	130k	R_0805_2012Metric	F	5.0%
R392	330k	R_0805_2012Metric	F	5.0%
R393	200k	R_0805_2012Metric	F	5.0%
R395	120k	R_0805_2012Metric	F	5.0%
R396	100k	R_0805_2012Metric	F	5.0%
R397	110k	R_0805_2012Metric	F	5.0%
R401	56k	R_0805_2012Metric	F	5.0%
R403	100k	R_0805_2012Metric	F	5.0%
R405	6.2M	R_0805_2012Metric	F	5.0%
R407	1.1M	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R413	18.2k	R_0805_2012Metric	F	1.0%
R414	29.4k	R_0805_2012Metric	F	1.0%
R415	14.7k	R_0805_2012Metric	F	1.0%
R416	22.6k	R_0805_2012Metric	F	1.0%
R417	100k	R_0805_2012Metric	F	1.0%
R418	100k	R_0805_2012Metric	F	1.0%
R419	100k	R_0805_2012Metric	F	1.0%
R420	100k	R_0805_2012Metric	F	1.0%
U1	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT

DNP component sites - leave pads open

Every reference below has value DNP, footprint R_0805_2012Metric (0805 resistor), side F. They are excluded from grouped fitted quantities and the combined shopping totals.

R001, R003, R004, R005, R006, R007, R010, R011, R014, R016, R017, R019, R022, R023, R024, R025, R028, R030, R031, R034, R036, R037, R039, R041, R043, R045, R047, R048, R049, R051, R054, R055, R057, R058, R061, R063, R065, R068, R069, R072, R073, R074, R075, R077, R078, R080, R081, R082, R083, R085, R088, R089, R091, R092, R096, R097, R099, R100, R101, R102.

R103, R105, R107, R108, R109, R110, R111, R113, R115, R117, R118, R120, R123, R124, R125, R129, R130, R134, R135, R139, R140, R143, R145, R146, R150, R151, R152, R155, R157, R158, R159, R160, R161, R163, R165, R166, R169, R171, R172, R175, R176, R178, R179, R182, R184, R185, R189, R190, R191, R195, R196, R201, R204, R206, R208, R211, R213, R215, R217, R219.

R220, R222, R225, R227, R229, R231, R232, R233, R234, R236, R238, R240, R242, R243, R244, R248, R249, R250, R251, R253, R255, R257, R258, R259, R262, R264, R267, R268, R270, R272, R273, R275, R278, R279, R281, R283, R284, R286, R287, R288, R290, R291, R292, R294, R297, R298, R299, R300, R302, R304, R305, R306, R307, R311, R316, R317, R321, R322, R324, R327.

R329, R330, R333, R334, R336, R338, R340, R343, R345, R347, R349, R350, R351, R353, R356, R357, R358, R359, R361, R362, R363, R364, R366, R368, R369, R370, R372, R373, R375, R378, R380, R382, R385, R387, R388, R389, R391, R394, R398, R399, R400, R402, R404, R406, R408, R409, R410, R411, R412, R421, R422, R423, R424, R425, R426, R427, R428, R429, R430, R431.

R432, R433, R434, R435, R436, R437, R438, R439, R440, R441, R442, R443, R444, R445, R446, R447, R448, R449, R450, R451, R452, R453, R454, R455, R456, R457, R458, R459, R460, R461, R462, R463, R464, R465, R466, R467, R468, R469, R470, R471, R472, R473, R474, R475, R476, R477, R478, R479, R480, R481, R482, R483, R484, R485, R486, R487, R488, R489, R490, R491.

R492, R493, R494, R495, R496, R497, R498, R499, R500, R501, R502, R503, R504, R505, R506, R507, R508, R509, R510, R511, R512, R513, R514, R515, R516, R517, R518, R519, R520, R521, R522, R523, R524, R525, R526, R527, R528, R529, R530, R531, R532.

DC1b

4 copper layers | 100 x 100 mm | 234 fitted items | 302 DNP sites

Populate ONLY variant DC1b for neurons N4 through N7. Same bare 4-layer board for all four variants. CSV overrides universal footprint values at all 532 resistor sites. There are 112 extra sites DNP on every variant; total DNP is larger. DNP means leave open, never fit 0 ohms.

Verified source: /home/kit/dc1-universal/fix12/board.kicad_pcb
SHA-256 b5d0b202d140751f72bbe6a50ee582ed0e2a69545f96704e8a4b415b7e787748
Population: /home/kit/dc1-universal/fix12/DC1b.csv | SHA-256 dc8f4575cfe6f9c0d7a2532a176f886542a23a639103ff16cf129b4cadc59bd9
Mounting holes (excluded from parts totals): H1 at (5, 5) mm; H2 at (95, 5) mm.

Grouped fitting summary

Value	Package	Qty	CSV tolerance
0.1uF	0805 capacitor	2	-
1.1M	0805 resistor	4	5.0%
1.3M	0805 resistor	2	5.0%
1.5M	0805 resistor	2	5.0%
1.6M	0805 resistor	2	5.0%
1.8M	0805 resistor	3	5.0%
100k	0805 resistor	4	1.0%
100k	0805 resistor	4	5.0%
10k	0805 resistor	1	1.0%
110k	0805 resistor	7	5.0%
120k	0805 resistor	5	5.0%
130k	0805 resistor	5	5.0%
13k	0805 resistor	1	1.0%
14.3k	0805 resistor	1	1.0%
150k	0805 resistor	9	5.0%
16.2k	0805 resistor	1	1.0%
160k	0805 resistor	10	5.0%
180k	0805 resistor	6	5.0%
1M	0805 resistor	3	5.0%
2.2M	0805 resistor	1	5.0%
200k	0805 resistor	14	5.0%
20k	0805 resistor	1	5.0%
220k	0805 resistor	6	5.0%

Value	Package	Qty	CSV tolerance
240k	0805 resistor	5	5.0%
24k	0805 resistor	1	5.0%
270k	0805 resistor	7	5.0%
27k	0805 resistor	1	5.0%
2M	0805 resistor	2	5.0%
300k	0805 resistor	12	5.0%
330k	0805 resistor	14	5.0%
33k	0805 resistor	2	5.0%
360k	0805 resistor	6	5.0%
390k	0805 resistor	5	5.0%
39k	0805 resistor	1	5.0%
430k	0805 resistor	5	5.0%
43k	0805 resistor	1	5.0%
470k	0805 resistor	8	5.0%
47k	0805 resistor	1	5.0%
510k	0805 resistor	11	5.0%
51k	0805 resistor	5	5.0%
560k	0805 resistor	5	5.0%
56k	0805 resistor	2	5.0%
620k	0805 resistor	1	5.0%
62k	0805 resistor	2	5.0%
680k	0805 resistor	7	5.0%
68k	0805 resistor	2	5.0%
750k	0805 resistor	5	5.0%
75k	0805 resistor	6	5.0%
820k	0805 resistor	4	5.0%
82k	0805 resistor	9	5.0%
910k	0805 resistor	1	5.0%
91k	0805 resistor	7	5.0%
PinHeader_2x38_Straddle	2x38 straddle, 2.54 mm	1	-
TL074	SOIC-14, 1.27 mm	1	-

Every populated component

Ref	Value	Footprint / package	Side	Tol / status
C_U1N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U1P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
J_STRADDLE_EDGE	PinHeader_2x38_Straddle	Connector_EdgeMount:PinHeader_2x38_Straddle_P2.54mm	F	FIT
R002	51k	R_0805_2012Metric	F	5.0%
R003	130k	R_0805_2012Metric	F	5.0%
R006	75k	R_0805_2012Metric	F	5.0%
R007	75k	R_0805_2012Metric	F	5.0%
R010	200k	R_0805_2012Metric	F	5.0%
R011	150k	R_0805_2012Metric	F	5.0%
R012	240k	R_0805_2012Metric	F	5.0%
R013	91k	R_0805_2012Metric	F	5.0%
R016	1.3M	R_0805_2012Metric	F	5.0%
R017	200k	R_0805_2012Metric	F	5.0%
R020	330k	R_0805_2012Metric	F	5.0%
R021	430k	R_0805_2012Metric	F	5.0%
R024	150k	R_0805_2012Metric	F	5.0%
R025	680k	R_0805_2012Metric	F	5.0%
R027	330k	R_0805_2012Metric	F	5.0%
R029	270k	R_0805_2012Metric	F	5.0%
R031	150k	R_0805_2012Metric	F	5.0%
R032	110k	R_0805_2012Metric	F	5.0%
R033	110k	R_0805_2012Metric	F	5.0%
R035	510k	R_0805_2012Metric	F	5.0%
R037	300k	R_0805_2012Metric	F	5.0%
R039	82k	R_0805_2012Metric	F	5.0%
R040	130k	R_0805_2012Metric	F	5.0%
R042	100k	R_0805_2012Metric	F	5.0%
R043	160k	R_0805_2012Metric	F	5.0%
R045	56k	R_0805_2012Metric	F	5.0%
R046	82k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R048	33k	R_0805_2012Metric	F	5.0%
R050	47k	R_0805_2012Metric	F	5.0%
R051	390k	R_0805_2012Metric	F	5.0%
R054	100k	R_0805_2012Metric	F	5.0%
R055	470k	R_0805_2012Metric	F	5.0%
R057	110k	R_0805_2012Metric	F	5.0%
R058	150k	R_0805_2012Metric	F	5.0%
R060	270k	R_0805_2012Metric	F	5.0%
R065	1M	R_0805_2012Metric	F	5.0%
R068	1.8M	R_0805_2012Metric	F	5.0%
R069	160k	R_0805_2012Metric	F	5.0%
R070	1.6M	R_0805_2012Metric	F	5.0%
R073	330k	R_0805_2012Metric	F	5.0%
R075	300k	R_0805_2012Metric	F	5.0%
R077	430k	R_0805_2012Metric	F	5.0%
R079	820k	R_0805_2012Metric	F	5.0%
R081	270k	R_0805_2012Metric	F	5.0%
R082	1.1M	R_0805_2012Metric	F	5.0%
R086	200k	R_0805_2012Metric	F	5.0%
R088	300k	R_0805_2012Metric	F	5.0%
R089	510k	R_0805_2012Metric	F	5.0%
R091	1.8M	R_0805_2012Metric	F	5.0%
R093	130k	R_0805_2012Metric	F	5.0%
R094	360k	R_0805_2012Metric	F	5.0%
R095	160k	R_0805_2012Metric	F	5.0%
R096	160k	R_0805_2012Metric	F	5.0%
R097	33k	R_0805_2012Metric	F	5.0%
R098	75k	R_0805_2012Metric	F	5.0%
R102	220k	R_0805_2012Metric	F	5.0%
R103	220k	R_0805_2012Metric	F	5.0%
R105	510k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R110	330k	R_0805_2012Metric	F	5.0%
R113	560k	R_0805_2012Metric	F	5.0%
R116	330k	R_0805_2012Metric	F	5.0%
R120	910k	R_0805_2012Metric	F	5.0%
R124	360k	R_0805_2012Metric	F	5.0%
R125	160k	R_0805_2012Metric	F	5.0%
R128	240k	R_0805_2012Metric	F	5.0%
R129	430k	R_0805_2012Metric	F	5.0%
R130	82k	R_0805_2012Metric	F	5.0%
R133	820k	R_0805_2012Metric	F	5.0%
R135	75k	R_0805_2012Metric	F	5.0%
R137	270k	R_0805_2012Metric	F	5.0%
R138	680k	R_0805_2012Metric	F	5.0%
R139	470k	R_0805_2012Metric	F	5.0%
R140	56k	R_0805_2012Metric	F	5.0%
R142	75k	R_0805_2012Metric	F	5.0%
R145	91k	R_0805_2012Metric	F	5.0%
R146	51k	R_0805_2012Metric	F	5.0%
R148	24k	R_0805_2012Metric	F	5.0%
R149	91k	R_0805_2012Metric	F	5.0%
R150	300k	R_0805_2012Metric	F	5.0%
R155	110k	R_0805_2012Metric	F	5.0%
R157	300k	R_0805_2012Metric	F	5.0%
R158	360k	R_0805_2012Metric	F	5.0%
R163	510k	R_0805_2012Metric	F	5.0%
R165	750k	R_0805_2012Metric	F	5.0%
R166	510k	R_0805_2012Metric	F	5.0%
R169	1.5M	R_0805_2012Metric	F	5.0%
R170	160k	R_0805_2012Metric	F	5.0%
R172	120k	R_0805_2012Metric	F	5.0%
R175	2M	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R176	220k	R_0805_2012Metric	F	5.0%
R178	180k	R_0805_2012Metric	F	5.0%
R180	200k	R_0805_2012Metric	F	5.0%
R181	220k	R_0805_2012Metric	F	5.0%
R182	300k	R_0805_2012Metric	F	5.0%
R184	1.3M	R_0805_2012Metric	F	5.0%
R186	330k	R_0805_2012Metric	F	5.0%
R187	62k	R_0805_2012Metric	F	5.0%
R188	220k	R_0805_2012Metric	F	5.0%
R189	430k	R_0805_2012Metric	F	5.0%
R191	150k	R_0805_2012Metric	F	5.0%
R193	20k	R_0805_2012Metric	F	5.0%
R195	82k	R_0805_2012Metric	F	5.0%
R196	100k	R_0805_2012Metric	F	5.0%
R198	68k	R_0805_2012Metric	F	5.0%
R199	43k	R_0805_2012Metric	F	5.0%
R200	82k	R_0805_2012Metric	F	5.0%
R201	91k	R_0805_2012Metric	F	5.0%
R203	560k	R_0805_2012Metric	F	5.0%
R206	160k	R_0805_2012Metric	F	5.0%
R207	360k	R_0805_2012Metric	F	5.0%
R209	91k	R_0805_2012Metric	F	5.0%
R210	120k	R_0805_2012Metric	F	5.0%
R212	680k	R_0805_2012Metric	F	5.0%
R213	390k	R_0805_2012Metric	F	5.0%
R215	180k	R_0805_2012Metric	F	5.0%
R216	680k	R_0805_2012Metric	F	5.0%
R219	1.1M	R_0805_2012Metric	F	5.0%
R220	200k	R_0805_2012Metric	F	5.0%
R223	300k	R_0805_2012Metric	F	5.0%
R225	390k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R226	430k	R_0805_2012Metric	F	5.0%
R227	150k	R_0805_2012Metric	F	5.0%
R229	470k	R_0805_2012Metric	F	5.0%
R231	560k	R_0805_2012Metric	F	5.0%
R233	750k	R_0805_2012Metric	F	5.0%
R234	470k	R_0805_2012Metric	F	5.0%
R236	620k	R_0805_2012Metric	F	5.0%
R239	1M	R_0805_2012Metric	F	5.0%
R240	270k	R_0805_2012Metric	F	5.0%
R241	130k	R_0805_2012Metric	F	5.0%
R242	330k	R_0805_2012Metric	F	5.0%
R245	330k	R_0805_2012Metric	F	5.0%
R246	220k	R_0805_2012Metric	F	5.0%
R247	330k	R_0805_2012Metric	F	5.0%
R250	300k	R_0805_2012Metric	F	5.0%
R252	68k	R_0805_2012Metric	F	5.0%
R253	27k	R_0805_2012Metric	F	5.0%
R255	51k	R_0805_2012Metric	F	5.0%
R257	120k	R_0805_2012Metric	F	5.0%
R259	110k	R_0805_2012Metric	F	5.0%
R260	110k	R_0805_2012Metric	F	5.0%
R261	51k	R_0805_2012Metric	F	5.0%
R263	150k	R_0805_2012Metric	F	5.0%
R264	470k	R_0805_2012Metric	F	5.0%
R266	82k	R_0805_2012Metric	F	5.0%
R268	82k	R_0805_2012Metric	F	5.0%
R269	470k	R_0805_2012Metric	F	5.0%
R270	510k	R_0805_2012Metric	F	5.0%
R271	100k	R_0805_2012Metric	F	5.0%
R273	110k	R_0805_2012Metric	F	5.0%
R274	330k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R276	680k	R_0805_2012Metric	F	5.0%
R277	390k	R_0805_2012Metric	F	5.0%
R279	180k	R_0805_2012Metric	F	5.0%
R280	510k	R_0805_2012Metric	F	5.0%
R282	300k	R_0805_2012Metric	F	5.0%
R283	300k	R_0805_2012Metric	F	5.0%
R285	240k	R_0805_2012Metric	F	5.0%
R288	240k	R_0805_2012Metric	F	5.0%
R289	510k	R_0805_2012Metric	F	5.0%
R290	200k	R_0805_2012Metric	F	5.0%
R291	820k	R_0805_2012Metric	F	5.0%
R293	300k	R_0805_2012Metric	F	5.0%
R294	680k	R_0805_2012Metric	F	5.0%
R297	360k	R_0805_2012Metric	F	5.0%
R299	82k	R_0805_2012Metric	F	5.0%
R301	200k	R_0805_2012Metric	F	5.0%
R302	82k	R_0805_2012Metric	F	5.0%
R305	91k	R_0805_2012Metric	F	5.0%
R306	150k	R_0805_2012Metric	F	5.0%
R307	510k	R_0805_2012Metric	F	5.0%
R311	200k	R_0805_2012Metric	F	5.0%
R313	510k	R_0805_2012Metric	F	5.0%
R314	1.1M	R_0805_2012Metric	F	5.0%
R316	180k	R_0805_2012Metric	F	5.0%
R318	470k	R_0805_2012Metric	F	5.0%
R321	120k	R_0805_2012Metric	F	5.0%
R322	200k	R_0805_2012Metric	F	5.0%
R325	2M	R_0805_2012Metric	F	5.0%
R327	2.2M	R_0805_2012Metric	F	5.0%
R329	200k	R_0805_2012Metric	F	5.0%
R330	750k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R334	750k	R_0805_2012Metric	F	5.0%
R336	360k	R_0805_2012Metric	F	5.0%
R339	300k	R_0805_2012Metric	F	5.0%
R342	130k	R_0805_2012Metric	F	5.0%
R343	510k	R_0805_2012Metric	F	5.0%
R345	470k	R_0805_2012Metric	F	5.0%
R348	150k	R_0805_2012Metric	F	5.0%
R350	160k	R_0805_2012Metric	F	5.0%
R353	91k	R_0805_2012Metric	F	5.0%
R357	39k	R_0805_2012Metric	F	5.0%
R358	160k	R_0805_2012Metric	F	5.0%
R362	200k	R_0805_2012Metric	F	5.0%
R364	1.5M	R_0805_2012Metric	F	5.0%
R365	560k	R_0805_2012Metric	F	5.0%
R367	160k	R_0805_2012Metric	F	5.0%
R368	180k	R_0805_2012Metric	F	5.0%
R370	270k	R_0805_2012Metric	F	5.0%
R371	1.8M	R_0805_2012Metric	F	5.0%
R373	820k	R_0805_2012Metric	F	5.0%
R374	330k	R_0805_2012Metric	F	5.0%
R375	750k	R_0805_2012Metric	F	5.0%
R376	680k	R_0805_2012Metric	F	5.0%
R378	240k	R_0805_2012Metric	F	5.0%
R379	200k	R_0805_2012Metric	F	5.0%
R380	1.6M	R_0805_2012Metric	F	5.0%
R383	200k	R_0805_2012Metric	F	5.0%
R385	560k	R_0805_2012Metric	F	5.0%
R387	120k	R_0805_2012Metric	F	5.0%
R389	330k	R_0805_2012Metric	F	5.0%
R391	270k	R_0805_2012Metric	F	5.0%
R394	1M	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R396	200k	R_0805_2012Metric	F	5.0%
R397	180k	R_0805_2012Metric	F	5.0%
R398	62k	R_0805_2012Metric	F	5.0%
R400	75k	R_0805_2012Metric	F	5.0%
R404	51k	R_0805_2012Metric	F	5.0%
R405	330k	R_0805_2012Metric	F	5.0%
R406	330k	R_0805_2012Metric	F	5.0%
R408	390k	R_0805_2012Metric	F	5.0%
R409	1.1M	R_0805_2012Metric	F	5.0%
R412	16.2k	R_0805_2012Metric	F	1.0%
R413	14.3k	R_0805_2012Metric	F	1.0%
R414	10k	R_0805_2012Metric	F	1.0%
R415	13k	R_0805_2012Metric	F	1.0%
R417	100k	R_0805_2012Metric	F	1.0%
R418	100k	R_0805_2012Metric	F	1.0%
R419	100k	R_0805_2012Metric	F	1.0%
R420	100k	R_0805_2012Metric	F	1.0%
U1	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT

DNP component sites - leave pads open

Every reference below has value DNP, footprint R_0805_2012Metric (0805 resistor), side F. They are excluded from grouped fitted quantities and the combined shopping totals.

R001, R004, R005, R008, R009, R014, R015, R018, R019, R022, R023, R026, R028, R030, R034, R036, R038, R041, R044, R047, R049, R052, R053, R056, R059, R061, R062, R063, R064, R066, R067, R071, R072, R074, R076, R078, R080, R083, R084, R085, R087, R090, R092, R099, R100, R101, R104, R106, R107, R108, R109, R111, R112, R114, R115, R117, R118, R119, R121, R122.

R123, R126, R127, R131, R132, R134, R136, R141, R143, R144, R147, R151, R152, R153, R154, R156, R159, R160, R161, R162, R164, R167, R168, R171, R173, R174, R177, R179, R183, R185, R190, R192, R194, R197, R202, R204, R205, R208, R211, R214, R217, R218, R221, R222, R224, R228, R230, R232, R235, R237, R238, R243, R244, R248, R249, R251, R254, R256, R258, R262.

R265, R267, R272, R275, R278, R281, R284, R286, R287, R292, R295, R296, R298, R300, R303, R304, R308, R309, R310, R312, R315, R317, R319, R320, R323, R324, R326, R328, R331, R332, R333, R335, R337, R338, R340, R341, R344, R346, R347, R349, R351, R352, R354, R355, R356, R359, R360, R361, R363, R366, R369, R372, R377, R381, R382, R384, R386, R388, R390, R392.

R393, R395, R399, R401, R402, R403, R407, R410, R411, R416, R421, R422, R423, R424, R425, R426, R427, R428, R429, R430, R431, R432, R433, R434, R435, R436, R437, R438, R439, R440, R441, R442, R443, R444, R445, R446, R447, R448, R449, R450, R451, R452, R453, R454, R455, R456, R457, R458, R459, R460, R461, R462, R463, R464, R465, R466, R467, R468, R469, R470.

R471, R472, R473, R474, R475, R476, R477, R478, R479, R480, R481, R482, R483, R484, R485, R486, R487, R488, R489, R490, R491, R492, R493, R494, R495, R496, R497, R498, R499, R500, R501, R502, R503, R504, R505, R506, R507, R508, R509, R510, R511, R512, R513, R514, R515, R516, R517, R518, R519, R520, R521, R522, R523, R524, R525, R526, R527, R528, R529, R530.

R531, R532.

DC1c

4 copper layers | 100 x 100 mm | 185 fitted items | 351 DNP sites

Populate ONLY variant DC1c for neurons N8 through N11. Same bare 4-layer board for all four variants. CSV overrides universal footprint values at all 532 resistor sites. There are 112 extra sites DNP on every variant; total DNP is larger. DNP means leave open, never fit 0 ohms.

Verified source: /home/kit/dc1-universal/fix12/board.kicad_pcb
SHA-256 b5d0b202d140751f72bbe6a50ee582ed0e2a69545f96704e8a4b415b7e787748
Population: /home/kit/dc1-universal/fix12/DC1c.csv | SHA-256 529908cd752bc10a99e0f43e793a51b9577f2a5a4433c3b0a5248588ddda1421
Mounting holes (excluded from parts totals): H1 at (5, 5) mm; H2 at (95, 5) mm.

Grouped fitting summary

Value	Package	Qty	CSV tolerance
0.1uF	0805 capacitor	2	-
1.1M	0805 resistor	2	5.0%
1.2M	0805 resistor	2	5.0%
1.3M	0805 resistor	2	5.0%
100k	0805 resistor	4	1.0%
100k	0805 resistor	2	5.0%
110k	0805 resistor	8	5.0%
120k	0805 resistor	3	5.0%
130k	0805 resistor	6	5.0%
150k	0805 resistor	5	5.0%
160k	0805 resistor	6	5.0%
18.2k	0805 resistor	1	1.0%
180k	0805 resistor	8	5.0%
1M	0805 resistor	2	5.0%
2.2M	0805 resistor	1	5.0%
200k	0805 resistor	4	5.0%
220k	0805 resistor	7	5.0%
240k	0805 resistor	8	5.0%
270k	0805 resistor	16	5.0%
27k	0805 resistor	1	5.0%
2M	0805 resistor	1	5.0%
300k	0805 resistor	7	5.0%
330k	0805 resistor	5	5.0%

Value	Package	Qty	CSV tolerance
33k	0805 resistor	1	5.0%
35.7k	0805 resistor	1	1.0%
360k	0805 resistor	5	5.0%
36k	0805 resistor	1	5.0%
390k	0805 resistor	4	5.0%
39k	0805 resistor	3	5.0%
3M	0805 resistor	1	5.0%
430k	0805 resistor	3	5.0%
43k	0805 resistor	2	5.0%
470k	0805 resistor	5	5.0%
47k	0805 resistor	4	5.0%
510k	0805 resistor	4	5.0%
51k	0805 resistor	1	5.0%
560k	0805 resistor	6	5.0%
56k	0805 resistor	2	5.0%
620k	0805 resistor	4	5.0%
62k	0805 resistor	4	5.0%
680k	0805 resistor	1	5.0%
68k	0805 resistor	3	5.0%
750k	0805 resistor	6	5.0%
75k	0805 resistor	3	5.0%
8.66k	0805 resistor	1	1.0%
820k	0805 resistor	5	5.0%
82k	0805 resistor	1	5.0%
90.9k	0805 resistor	1	1.0%
910k	0805 resistor	3	5.0%
91k	0805 resistor	5	5.0%
PinHeader_2x38_Straddle	2x38 straddle, 2.54 mm	1	-
TL074	SOIC-14, 1.27 mm	1	-

Every populated component

Ref	Value	Footprint / package	Side	Tol / status
C_U1N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U1P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
J_STRADDLE_EDGE	PinHeader_2x38_Straddle	Connector_EdgeMount:PinHeader_2x38_Straddle_P2.54mm	F	FIT
R004	51k	R_0805_2012Metric	F	5.0%
R010	180k	R_0805_2012Metric	F	5.0%
R011	33k	R_0805_2012Metric	F	5.0%
R012	43k	R_0805_2012Metric	F	5.0%
R016	270k	R_0805_2012Metric	F	5.0%
R019	62k	R_0805_2012Metric	F	5.0%
R022	1M	R_0805_2012Metric	F	5.0%
R023	62k	R_0805_2012Metric	F	5.0%
R026	120k	R_0805_2012Metric	F	5.0%
R029	180k	R_0805_2012Metric	F	5.0%
R030	43k	R_0805_2012Metric	F	5.0%
R032	160k	R_0805_2012Metric	F	5.0%
R035	360k	R_0805_2012Metric	F	5.0%
R036	75k	R_0805_2012Metric	F	5.0%
R039	240k	R_0805_2012Metric	F	5.0%
R042	330k	R_0805_2012Metric	F	5.0%
R043	110k	R_0805_2012Metric	F	5.0%
R047	47k	R_0805_2012Metric	F	5.0%
R048	39k	R_0805_2012Metric	F	5.0%
R049	27k	R_0805_2012Metric	F	5.0%
R054	75k	R_0805_2012Metric	F	5.0%
R056	91k	R_0805_2012Metric	F	5.0%
R057	39k	R_0805_2012Metric	F	5.0%
R061	620k	R_0805_2012Metric	F	5.0%
R063	360k	R_0805_2012Metric	F	5.0%
R065	100k	R_0805_2012Metric	F	5.0%
R068	220k	R_0805_2012Metric	F	5.0%
R069	240k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R070	91k	R_0805_2012Metric	F	5.0%
R073	360k	R_0805_2012Metric	F	5.0%
R074	180k	R_0805_2012Metric	F	5.0%
R077	110k	R_0805_2012Metric	F	5.0%
R079	430k	R_0805_2012Metric	F	5.0%
R080	300k	R_0805_2012Metric	F	5.0%
R081	330k	R_0805_2012Metric	F	5.0%
R084	220k	R_0805_2012Metric	F	5.0%
R085	180k	R_0805_2012Metric	F	5.0%
R086	560k	R_0805_2012Metric	F	5.0%
R090	510k	R_0805_2012Metric	F	5.0%
R092	160k	R_0805_2012Metric	F	5.0%
R095	68k	R_0805_2012Metric	F	5.0%
R096	110k	R_0805_2012Metric	F	5.0%
R097	82k	R_0805_2012Metric	F	5.0%
R100	110k	R_0805_2012Metric	F	5.0%
R101	91k	R_0805_2012Metric	F	5.0%
R103	47k	R_0805_2012Metric	F	5.0%
R108	160k	R_0805_2012Metric	F	5.0%
R109	220k	R_0805_2012Metric	F	5.0%
R111	47k	R_0805_2012Metric	F	5.0%
R116	120k	R_0805_2012Metric	F	5.0%
R117	160k	R_0805_2012Metric	F	5.0%
R118	62k	R_0805_2012Metric	F	5.0%
R123	330k	R_0805_2012Metric	F	5.0%
R125	75k	R_0805_2012Metric	F	5.0%
R128	470k	R_0805_2012Metric	F	5.0%
R129	390k	R_0805_2012Metric	F	5.0%
R133	130k	R_0805_2012Metric	F	5.0%
R134	240k	R_0805_2012Metric	F	5.0%
R136	910k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R138	120k	R_0805_2012Metric	F	5.0%
R141	270k	R_0805_2012Metric	F	5.0%
R144	130k	R_0805_2012Metric	F	5.0%
R147	180k	R_0805_2012Metric	F	5.0%
R149	200k	R_0805_2012Metric	F	5.0%
R151	130k	R_0805_2012Metric	F	5.0%
R152	56k	R_0805_2012Metric	F	5.0%
R154	750k	R_0805_2012Metric	F	5.0%
R157	470k	R_0805_2012Metric	F	5.0%
R159	150k	R_0805_2012Metric	F	5.0%
R160	750k	R_0805_2012Metric	F	5.0%
R165	560k	R_0805_2012Metric	F	5.0%
R166	270k	R_0805_2012Metric	F	5.0%
R169	2.2M	R_0805_2012Metric	F	5.0%
R170	470k	R_0805_2012Metric	F	5.0%
R173	820k	R_0805_2012Metric	F	5.0%
R178	240k	R_0805_2012Metric	F	5.0%
R179	270k	R_0805_2012Metric	F	5.0%
R181	180k	R_0805_2012Metric	F	5.0%
R183	1.2M	R_0805_2012Metric	F	5.0%
R184	150k	R_0805_2012Metric	F	5.0%
R185	1.3M	R_0805_2012Metric	F	5.0%
R188	200k	R_0805_2012Metric	F	5.0%
R190	130k	R_0805_2012Metric	F	5.0%
R196	3M	R_0805_2012Metric	F	5.0%
R200	110k	R_0805_2012Metric	F	5.0%
R201	91k	R_0805_2012Metric	F	5.0%
R204	47k	R_0805_2012Metric	F	5.0%
R205	910k	R_0805_2012Metric	F	5.0%
R207	430k	R_0805_2012Metric	F	5.0%
R209	270k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R211	330k	R_0805_2012Metric	F	5.0%
R215	160k	R_0805_2012Metric	F	5.0%
R216	910k	R_0805_2012Metric	F	5.0%
R221	390k	R_0805_2012Metric	F	5.0%
R225	750k	R_0805_2012Metric	F	5.0%
R228	750k	R_0805_2012Metric	F	5.0%
R229	1.1M	R_0805_2012Metric	F	5.0%
R231	1M	R_0805_2012Metric	F	5.0%
R234	270k	R_0805_2012Metric	F	5.0%
R238	270k	R_0805_2012Metric	F	5.0%
R241	91k	R_0805_2012Metric	F	5.0%
R243	510k	R_0805_2012Metric	F	5.0%
R244	510k	R_0805_2012Metric	F	5.0%
R247	240k	R_0805_2012Metric	F	5.0%
R249	130k	R_0805_2012Metric	F	5.0%
R251	270k	R_0805_2012Metric	F	5.0%
R257	68k	R_0805_2012Metric	F	5.0%
R261	240k	R_0805_2012Metric	F	5.0%
R264	390k	R_0805_2012Metric	F	5.0%
R266	470k	R_0805_2012Metric	F	5.0%
R268	300k	R_0805_2012Metric	F	5.0%
R271	300k	R_0805_2012Metric	F	5.0%
R273	180k	R_0805_2012Metric	F	5.0%
R274	560k	R_0805_2012Metric	F	5.0%
R277	270k	R_0805_2012Metric	F	5.0%
R279	270k	R_0805_2012Metric	F	5.0%
R283	180k	R_0805_2012Metric	F	5.0%
R285	270k	R_0805_2012Metric	F	5.0%
R287	2M	R_0805_2012Metric	F	5.0%
R289	240k	R_0805_2012Metric	F	5.0%
R290	150k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R291	270k	R_0805_2012Metric	F	5.0%
R292	430k	R_0805_2012Metric	F	5.0%
R296	560k	R_0805_2012Metric	F	5.0%
R298	39k	R_0805_2012Metric	F	5.0%
R301	360k	R_0805_2012Metric	F	5.0%
R304	1.1M	R_0805_2012Metric	F	5.0%
R308	270k	R_0805_2012Metric	F	5.0%
R312	270k	R_0805_2012Metric	F	5.0%
R315	220k	R_0805_2012Metric	F	5.0%
R320	220k	R_0805_2012Metric	F	5.0%
R322	750k	R_0805_2012Metric	F	5.0%
R328	620k	R_0805_2012Metric	F	5.0%
R331	750k	R_0805_2012Metric	F	5.0%
R335	620k	R_0805_2012Metric	F	5.0%
R338	820k	R_0805_2012Metric	F	5.0%
R340	620k	R_0805_2012Metric	F	5.0%
R341	300k	R_0805_2012Metric	F	5.0%
R343	820k	R_0805_2012Metric	F	5.0%
R345	200k	R_0805_2012Metric	F	5.0%
R347	1.2M	R_0805_2012Metric	F	5.0%
R349	56k	R_0805_2012Metric	F	5.0%
R351	1.3M	R_0805_2012Metric	F	5.0%
R355	62k	R_0805_2012Metric	F	5.0%
R356	110k	R_0805_2012Metric	F	5.0%
R359	100k	R_0805_2012Metric	F	5.0%
R361	300k	R_0805_2012Metric	F	5.0%
R363	130k	R_0805_2012Metric	F	5.0%
R366	330k	R_0805_2012Metric	F	5.0%
R367	150k	R_0805_2012Metric	F	5.0%
R368	680k	R_0805_2012Metric	F	5.0%
R369	220k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R371	220k	R_0805_2012Metric	F	5.0%
R372	560k	R_0805_2012Metric	F	5.0%
R374	360k	R_0805_2012Metric	F	5.0%
R375	390k	R_0805_2012Metric	F	5.0%
R376	270k	R_0805_2012Metric	F	5.0%
R377	300k	R_0805_2012Metric	F	5.0%
R379	510k	R_0805_2012Metric	F	5.0%
R380	470k	R_0805_2012Metric	F	5.0%
R382	820k	R_0805_2012Metric	F	5.0%
R384	110k	R_0805_2012Metric	F	5.0%
R387	300k	R_0805_2012Metric	F	5.0%
R388	68k	R_0805_2012Metric	F	5.0%
R394	110k	R_0805_2012Metric	F	5.0%
R396	200k	R_0805_2012Metric	F	5.0%
R397	150k	R_0805_2012Metric	F	5.0%
R398	240k	R_0805_2012Metric	F	5.0%
R399	160k	R_0805_2012Metric	F	5.0%
R401	36k	R_0805_2012Metric	F	5.0%
R404	560k	R_0805_2012Metric	F	5.0%
R405	820k	R_0805_2012Metric	F	5.0%
R406	270k	R_0805_2012Metric	F	5.0%
R410	18.2k	R_0805_2012Metric	F	1.0%
R413	8.66k	R_0805_2012Metric	F	1.0%
R415	35.7k	R_0805_2012Metric	F	1.0%
R416	90.9k	R_0805_2012Metric	F	1.0%
R417	100k	R_0805_2012Metric	F	1.0%
R418	100k	R_0805_2012Metric	F	1.0%
R419	100k	R_0805_2012Metric	F	1.0%
R420	100k	R_0805_2012Metric	F	1.0%
U1	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT

DNP component sites - leave pads open

Every reference below has value DNP, footprint R_0805_2012Metric (0805 resistor), side F. They are excluded from grouped fitted quantities and the combined shopping totals.

R001, R002, R003, R005, R006, R007, R008, R009, R013, R014, R015, R017, R018, R020, R021, R024, R025, R027, R028, R031, R033, R034, R037, R038, R040, R041, R044, R045, R046, R050, R051, R052, R053, R055, R058, R059, R060, R062, R064, R066, R067, R071, R072, R075, R076, R078, R082, R083, R087, R088, R089, R091, R093, R094, R098, R099, R102, R104, R105, R106.

R107, R110, R112, R113, R114, R115, R119, R120, R121, R122, R124, R126, R127, R130, R131, R132, R135, R137, R139, R140, R142, R143, R145, R146, R148, R150, R153, R155, R156, R158, R161, R162, R163, R164, R167, R168, R171, R172, R174, R175, R176, R177, R180, R182, R186, R187, R189, R191, R192, R193, R194, R195, R197, R198, R199, R202, R203, R206, R208, R210.

R212, R213, R214, R217, R218, R219, R220, R222, R223, R224, R226, R227, R230, R232, R233, R235, R236, R237, R239, R240, R242, R245, R246, R248, R250, R252, R253, R254, R255, R256, R258, R259, R260, R262, R263, R265, R267, R269, R270, R272, R275, R276, R278, R280, R281, R282, R284, R286, R288, R293, R294, R295, R297, R299, R300, R302, R303, R305, R306, R307.

R309, R310, R311, R313, R314, R316, R317, R318, R319, R321, R323, R324, R325, R326, R327, R329, R330, R332, R333, R334, R336, R337, R339, R342, R344, R346, R348, R350, R352, R353, R354, R357, R358, R360, R362, R364, R365, R370, R373, R378, R381, R383, R385, R386, R389, R390, R391, R392, R393, R395, R400, R402, R403, R407, R408, R409, R411, R412, R414, R421.

R422, R423, R424, R425, R426, R427, R428, R429, R430, R431, R432, R433, R434, R435, R436, R437, R438, R439, R440, R441, R442, R443, R444, R445, R446, R447, R448, R449, R450, R451, R452, R453, R454, R455, R456, R457, R458, R459, R460, R461, R462, R463, R464, R465, R466, R467, R468, R469, R470, R471, R472, R473, R474, R475, R476, R477, R478, R479, R480, R481.

R482, R483, R484, R485, R486, R487, R488, R489, R490, R491, R492, R493, R494, R495, R496, R497, R498, R499, R500, R501, R502, R503, R504, R505, R506, R507, R508, R509, R510, R511, R512, R513, R514, R515, R516, R517, R518, R519, R520, R521, R522, R523, R524, R525, R526, R527, R528, R529, R530, R531, R532.

DC1d

4 copper layers | 100 x 100 mm | 145 fitted items | 391 DNP sites

Populate ONLY variant DC1d for neurons N12 through N15. Same bare 4-layer board for all four variants. CSV overrides universal footprint values at all 532 resistor sites. There are 112 extra sites DNP on every variant; total DNP is larger. DNP means leave open, never fit 0 ohms.

Verified source: /home/kit/dc1-universal/fix12/board.kicad_pcb
SHA-256 b5d0b202d140751f72bbe6a50ee582ed0e2a69545f96704e8a4b415b7e787748
Population: /home/kit/dc1-universal/fix12/DC1d.csv | SHA-256 9f6b2f55462228f263b343a11189928661a399a0965fbb7a27cb3b58a6a65ca9
Mounting holes (excluded from parts totals): H1 at (5, 5) mm; H2 at (95, 5) mm.

Grouped fitting summary

Value	Package	Qty	CSV tolerance
0.1uF	0805 capacitor	2	-
1.1M	0805 resistor	1	5.0%
1.2M	0805 resistor	1	5.0%
1.3M	0805 resistor	2	5.0%
1.5M	0805 resistor	1	5.0%
100k	0805 resistor	4	1.0%
100k	0805 resistor	4	5.0%
110k	0805 resistor	2	5.0%
113k	0805 resistor	1	1.0%
120k	0805 resistor	3	5.0%
130k	0805 resistor	4	5.0%
15.4k	0805 resistor	1	1.0%
150k	0805 resistor	6	5.0%
160k	0805 resistor	11	5.0%
180k	0805 resistor	6	5.0%
1M	0805 resistor	1	5.0%
2.2M	0805 resistor	1	5.0%
200k	0805 resistor	8	5.0%
220k	0805 resistor	7	5.0%
240k	0805 resistor	4	5.0%
270k	0805 resistor	9	5.0%
2M	0805 resistor	1	5.0%
300k	0805 resistor	5	5.0%

Value	Package	Qty	CSV tolerance
330k	0805 resistor	8	5.0%
360k	0805 resistor	5	5.0%
36k	0805 resistor	2	5.0%
390k	0805 resistor	4	5.0%
39k	0805 resistor	2	5.0%
430k	0805 resistor	7	5.0%
43k	0805 resistor	1	5.0%
470k	0805 resistor	3	5.0%
510k	0805 resistor	2	5.0%
51k	0805 resistor	1	5.0%
52.3k	0805 resistor	1	1.0%
560k	0805 resistor	2	5.0%
61.9k	0805 resistor	1	1.0%
62k	0805 resistor	2	5.0%
680k	0805 resistor	1	5.0%
68k	0805 resistor	2	5.0%
750k	0805 resistor	4	5.0%
75k	0805 resistor	3	5.0%
820k	0805 resistor	1	5.0%
82k	0805 resistor	2	5.0%
910k	0805 resistor	2	5.0%
91k	0805 resistor	2	5.0%
PinHeader_2x38_Straddle	2x38 straddle, 2.54 mm	1	-
TL074	SOIC-14, 1.27 mm	1	-

Every populated component

Ref	Value	Footprint / package	Side	Tol / status
C_U1N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U1P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
J_STRADDLE_EDGE	PinHeader_2x38_Straddle	Connector_EdgeMount:PinHeader_2x38_Straddle_P2.54mm	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
R001	43k	R_0805_2012Metric	F	5.0%
R005	39k	R_0805_2012Metric	F	5.0%
R009	200k	R_0805_2012Metric	F	5.0%
R012	91k	R_0805_2012Metric	F	5.0%
R014	220k	R_0805_2012Metric	F	5.0%
R015	390k	R_0805_2012Metric	F	5.0%
R019	100k	R_0805_2012Metric	F	5.0%
R021	200k	R_0805_2012Metric	F	5.0%
R026	150k	R_0805_2012Metric	F	5.0%
R028	300k	R_0805_2012Metric	F	5.0%
R034	270k	R_0805_2012Metric	F	5.0%
R039	200k	R_0805_2012Metric	F	5.0%
R041	750k	R_0805_2012Metric	F	5.0%
R045	110k	R_0805_2012Metric	F	5.0%
R047	270k	R_0805_2012Metric	F	5.0%
R054	160k	R_0805_2012Metric	F	5.0%
R056	430k	R_0805_2012Metric	F	5.0%
R057	36k	R_0805_2012Metric	F	5.0%
R059	180k	R_0805_2012Metric	F	5.0%
R060	910k	R_0805_2012Metric	F	5.0%
R065	270k	R_0805_2012Metric	F	5.0%
R067	300k	R_0805_2012Metric	F	5.0%
R070	360k	R_0805_2012Metric	F	5.0%
R072	270k	R_0805_2012Metric	F	5.0%
R077	330k	R_0805_2012Metric	F	5.0%
R078	360k	R_0805_2012Metric	F	5.0%
R080	750k	R_0805_2012Metric	F	5.0%
R081	1.1M	R_0805_2012Metric	F	5.0%
R082	220k	R_0805_2012Metric	F	5.0%
R083	390k	R_0805_2012Metric	F	5.0%
R086	270k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R089	330k	R_0805_2012Metric	F	5.0%
R093	300k	R_0805_2012Metric	F	5.0%
R095	220k	R_0805_2012Metric	F	5.0%
R097	120k	R_0805_2012Metric	F	5.0%
R099	330k	R_0805_2012Metric	F	5.0%
R103	75k	R_0805_2012Metric	F	5.0%
R107	510k	R_0805_2012Metric	F	5.0%
R111	220k	R_0805_2012Metric	F	5.0%
R115	1M	R_0805_2012Metric	F	5.0%
R119	750k	R_0805_2012Metric	F	5.0%
R122	130k	R_0805_2012Metric	F	5.0%
R128	110k	R_0805_2012Metric	F	5.0%
R131	820k	R_0805_2012Metric	F	5.0%
R136	2M	R_0805_2012Metric	F	5.0%
R138	1.5M	R_0805_2012Metric	F	5.0%
R143	510k	R_0805_2012Metric	F	5.0%
R146	51k	R_0805_2012Metric	F	5.0%
R149	220k	R_0805_2012Metric	F	5.0%
R153	150k	R_0805_2012Metric	F	5.0%
R157	390k	R_0805_2012Metric	F	5.0%
R161	160k	R_0805_2012Metric	F	5.0%
R165	360k	R_0805_2012Metric	F	5.0%
R167	150k	R_0805_2012Metric	F	5.0%
R171	360k	R_0805_2012Metric	F	5.0%
R173	1.3M	R_0805_2012Metric	F	5.0%
R176	560k	R_0805_2012Metric	F	5.0%
R180	430k	R_0805_2012Metric	F	5.0%
R182	330k	R_0805_2012Metric	F	5.0%
R186	470k	R_0805_2012Metric	F	5.0%
R194	240k	R_0805_2012Metric	F	5.0%
R198	39k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R200	270k	R_0805_2012Metric	F	5.0%
R204	470k	R_0805_2012Metric	F	5.0%
R208	330k	R_0805_2012Metric	F	5.0%
R211	160k	R_0805_2012Metric	F	5.0%
R214	220k	R_0805_2012Metric	F	5.0%
R217	300k	R_0805_2012Metric	F	5.0%
R221	180k	R_0805_2012Metric	F	5.0%
R222	130k	R_0805_2012Metric	F	5.0%
R225	360k	R_0805_2012Metric	F	5.0%
R228	330k	R_0805_2012Metric	F	5.0%
R231	160k	R_0805_2012Metric	F	5.0%
R232	160k	R_0805_2012Metric	F	5.0%
R234	430k	R_0805_2012Metric	F	5.0%
R239	270k	R_0805_2012Metric	F	5.0%
R241	220k	R_0805_2012Metric	F	5.0%
R245	240k	R_0805_2012Metric	F	5.0%
R248	130k	R_0805_2012Metric	F	5.0%
R251	91k	R_0805_2012Metric	F	5.0%
R256	430k	R_0805_2012Metric	F	5.0%
R258	680k	R_0805_2012Metric	F	5.0%
R262	180k	R_0805_2012Metric	F	5.0%
R264	160k	R_0805_2012Metric	F	5.0%
R267	470k	R_0805_2012Metric	F	5.0%
R269	270k	R_0805_2012Metric	F	5.0%
R272	2.2M	R_0805_2012Metric	F	5.0%
R275	910k	R_0805_2012Metric	F	5.0%
R278	430k	R_0805_2012Metric	F	5.0%
R281	560k	R_0805_2012Metric	F	5.0%
R284	430k	R_0805_2012Metric	F	5.0%
R286	180k	R_0805_2012Metric	F	5.0%
R288	1.3M	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R296	120k	R_0805_2012Metric	F	5.0%
R300	100k	R_0805_2012Metric	F	5.0%
R305	130k	R_0805_2012Metric	F	5.0%
R307	150k	R_0805_2012Metric	F	5.0%
R311	270k	R_0805_2012Metric	F	5.0%
R313	68k	R_0805_2012Metric	F	5.0%
R316	300k	R_0805_2012Metric	F	5.0%
R317	82k	R_0805_2012Metric	F	5.0%
R319	240k	R_0805_2012Metric	F	5.0%
R321	200k	R_0805_2012Metric	F	5.0%
R324	62k	R_0805_2012Metric	F	5.0%
R329	150k	R_0805_2012Metric	F	5.0%
R331	330k	R_0805_2012Metric	F	5.0%
R332	75k	R_0805_2012Metric	F	5.0%
R333	160k	R_0805_2012Metric	F	5.0%
R336	120k	R_0805_2012Metric	F	5.0%
R339	62k	R_0805_2012Metric	F	5.0%
R342	75k	R_0805_2012Metric	F	5.0%
R344	68k	R_0805_2012Metric	F	5.0%
R348	100k	R_0805_2012Metric	F	5.0%
R351	240k	R_0805_2012Metric	F	5.0%
R357	200k	R_0805_2012Metric	F	5.0%
R359	36k	R_0805_2012Metric	F	5.0%
R362	160k	R_0805_2012Metric	F	5.0%
R366	82k	R_0805_2012Metric	F	5.0%
R368	160k	R_0805_2012Metric	F	5.0%
R371	430k	R_0805_2012Metric	F	5.0%
R374	150k	R_0805_2012Metric	F	5.0%
R375	180k	R_0805_2012Metric	F	5.0%
R379	1.2M	R_0805_2012Metric	F	5.0%
R382	180k	R_0805_2012Metric	F	5.0%

Ref	Value	Footprint / package	Side	Tol / status
R387	160k	R_0805_2012Metric	F	5.0%
R393	200k	R_0805_2012Metric	F	5.0%
R395	200k	R_0805_2012Metric	F	5.0%
R396	330k	R_0805_2012Metric	F	5.0%
R398	100k	R_0805_2012Metric	F	5.0%
R400	390k	R_0805_2012Metric	F	5.0%
R402	160k	R_0805_2012Metric	F	5.0%
R405	750k	R_0805_2012Metric	F	5.0%
R407	200k	R_0805_2012Metric	F	5.0%
R411	113k	R_0805_2012Metric	F	1.0%
R413	15.4k	R_0805_2012Metric	F	1.0%
R414	52.3k	R_0805_2012Metric	F	1.0%
R416	61.9k	R_0805_2012Metric	F	1.0%
R417	100k	R_0805_2012Metric	F	1.0%
R418	100k	R_0805_2012Metric	F	1.0%
R419	100k	R_0805_2012Metric	F	1.0%
R420	100k	R_0805_2012Metric	F	1.0%
U1	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT

DNP component sites - leave pads open

Every reference below has value DNP, footprint R_0805_2012Metric (0805 resistor), side F. They are excluded from grouped fitted quantities and the combined shopping totals.

R002, R003, R004, R006, R007, R008, R010, R011, R013, R016, R017, R018, R020, R022, R023, R024, R025, R027, R029, R030, R031, R032, R033, R035, R036, R037, R038, R040, R042, R043, R044, R046, R048, R049, R050, R051, R052, R053, R055, R058, R061, R062, R063, R064, R066, R068, R069, R071, R073, R074, R075, R076, R079, R084, R085, R087, R088, R090, R091, R092.

R094, R096, R098, R100, R101, R102, R104, R105, R106, R108, R109, R110, R112, R113, R114, R116, R117, R118, R120, R121, R123, R124, R125, R126, R127, R129, R130, R132, R133, R134, R135, R137, R139, R140, R141, R142, R144, R145, R147, R148, R150, R151, R152, R154, R155, R156, R158, R159, R160, R162, R163, R164, R166, R168, R169, R170, R172, R174, R175, R177.

R178, R179, R181, R183, R184, R185, R187, R188, R189, R190, R191, R192, R193, R195, R196, R197, R199, R201, R202, R203, R205, R206, R207, R209, R210, R212, R213, R215, R216, R218, R219, R220, R223, R224, R226, R227, R229, R230, R233, R235, R236, R237, R238, R240, R242, R243, R244, R246, R247, R249, R250, R252, R253, R254, R255, R257, R259, R260, R261, R263.

R265, R266, R268, R270, R271, R273, R274, R276, R277, R279, R280, R282, R283, R285, R287, R289, R290, R291, R292, R293, R294, R295, R297, R298, R299, R301, R302, R303, R304, R306, R308, R309, R310, R312, R314, R315, R318, R320, R322, R323, R325, R326, R327, R328, R330, R334, R335, R337, R338, R340, R341, R343, R345, R346, R347, R349, R350, R352, R353, R354.

R355, R356, R358, R360, R361, R363, R364, R365, R367, R369, R370, R372, R373, R376, R377, R378, R380, R381, R383, R384, R385, R386, R388, R389, R390, R391, R392, R394, R397, R399, R401, R403, R404, R406, R408, R409, R410, R412, R415, R421, R422, R423, R424, R425, R426, R427, R428, R429, R430, R431, R432, R433, R434, R435, R436, R437, R438, R439, R440, R441.

R442, R443, R444, R445, R446, R447, R448, R449, R450, R451, R452, R453, R454, R455, R456, R457, R458, R459, R460, R461, R462, R463, R464, R465, R466, R467, R468, R469, R470, R471, R472, R473, R474, R475, R476, R477, R478, R479, R480, R481, R482, R483, R484, R485, R486, R487, R488, R489, R490, R491, R492, R493, R494, R495, R496, R497, R498, R499, R500, R501.

R502, R503, R504, R505, R506, R507, R508, R509, R510, R511, R512, R513, R514, R515, R516, R517, R518, R519, R520, R521, R522, R523, R524, R525, R526, R527, R528, R529, R530, R531, R532.

DC2

2 copper layers | 100 x 100 mm | 89 fitted items | 0 DNP sites

Verified source: /work/dc1_4l_emu/fresh_main/two_layer/daughtercard2_relu_l1_buffered_repaired.kicad_pcb (copied read-only to source_snapshots/DC2.kicad_pcb)
SHA-256 04167d23083c5d8512eeee996369d7a1ff4c091298bac89ebe6271c0bf2f0ce0
Mounting holes (excluded from parts totals): H1 at (5, 5) mm; H2 at (95, 5) mm.

Grouped fitting summary

Value	Package	Qty	CSV tolerance
0.1uF	0805 capacitor	16	-
10k	0805 resistor	32	-
PinHeader_2x38_Straddle	2x38 straddle, 2.54 mm	1	-
1N4148W	SOD-123	32	-
TL074	SOIC-14, 1.27 mm	8	-

Every populated component

Ref	Value	Footprint / package	Side	Tol / status
C_U1N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U1P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U2N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U2P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U3N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U3P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U4N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U4P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U_BUF1N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U_BUF1P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U_BUF2N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U_BUF2P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U_BUF3N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U_BUF3P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U_BUF4N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U_BUF4P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
D_CLAMP_0	1N4148W	D_SOD-123	F	FIT
D_CLAMP_1	1N4148W	D_SOD-123	F	FIT
D_CLAMP_2	1N4148W	D_SOD-123	F	FIT
D_CLAMP_3	1N4148W	D_SOD-123	F	FIT
D_CLAMP_4	1N4148W	D_SOD-123	F	FIT
D_CLAMP_5	1N4148W	D_SOD-123	F	FIT
D_CLAMP_6	1N4148W	D_SOD-123	F	FIT
D_CLAMP_7	1N4148W	D_SOD-123	F	FIT
D_CLAMP_8	1N4148W	D_SOD-123	F	FIT
D_CLAMP_9	1N4148W	D_SOD-123	F	FIT
D_CLAMP_10	1N4148W	D_SOD-123	F	FIT
D_CLAMP_11	1N4148W	D_SOD-123	F	FIT
D_CLAMP_12	1N4148W	D_SOD-123	F	FIT
D_CLAMP_13	1N4148W	D_SOD-123	F	FIT
D_CLAMP_14	1N4148W	D_SOD-123	F	FIT
D_CLAMP_15	1N4148W	D_SOD-123	F	FIT
D_RECT_0	1N4148W	D_SOD-123	F	FIT
D_RECT_1	1N4148W	D_SOD-123	F	FIT
D_RECT_2	1N4148W	D_SOD-123	F	FIT
D_RECT_3	1N4148W	D_SOD-123	F	FIT
D_RECT_4	1N4148W	D_SOD-123	F	FIT
D_RECT_5	1N4148W	D_SOD-123	F	FIT
D_RECT_6	1N4148W	D_SOD-123	F	FIT
D_RECT_7	1N4148W	D_SOD-123	F	FIT
D_RECT_8	1N4148W	D_SOD-123	F	FIT
D_RECT_9	1N4148W	D_SOD-123	F	FIT
D_RECT_10	1N4148W	D_SOD-123	F	FIT
D_RECT_11	1N4148W	D_SOD-123	F	FIT
D_RECT_12	1N4148W	D_SOD-123	F	FIT
D_RECT_13	1N4148W	D_SOD-123	F	FIT
D_RECT_14	1N4148W	D_SOD-123	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
D_RECT_15	1N4148W	D_SOD-123	F	FIT
J_STRADDLE_EDGE	PinHeader_2x38_Straddle	Connector_EdgeMount:PinHeader_2x38_Straddle_P2.54mm	F	FIT
R_FB_0	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_1	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_2	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_3	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_4	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_5	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_6	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_7	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_8	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_9	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_10	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_11	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_12	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_13	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_14	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_FB_15	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_0	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_1	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_2	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_3	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_4	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_5	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_6	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_7	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_8	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_9	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_10	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_11	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_12	10k	Resistor_SMD:R_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
R_IN_13	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_14	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
R_IN_15	10k	Resistor_SMD:R_0805_2012Metric	F	FIT
U1	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT
U2	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT
U3	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT
U4	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT
U_BUF1	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT
U_BUF2	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT
U_BUF3	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT
U_BUF4	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT

DC3_v2

2 copper layers | 100 x 100 mm | 163 fitted items | 217 DNP sites

Universal DC3 has 217 DNP resistor sites. Keep those sites open. Populated values and DNP state are taken directly from this final board.

Verified source: /home/kit/dc3-holes/dc3_universal.kicad_pcb
SHA-256 d7c2393ffd6dec2fd1368113ee96d8bb0d8e9bd686b861ff6524de71649526eb
Mounting holes (excluded from parts totals): H1 at (5, 5) mm; H2 at (95, 5) mm.

Grouped fitting summary

Value	Package	Qty	CSV tolerance
0.1uF	0805 capacitor	6	-
1.1M	0805 resistor	1	-
1.6M	0805 resistor	1	-
100k	0805 resistor	16	-
110k	0805 resistor	3	-
120k	0805 resistor	3	-
130k	0805 resistor	4	-
14.3k	0805 resistor	1	-
14.7k	0805 resistor	1	-
150k	0805 resistor	6	-
160k	0805 resistor	5	-
17.4k	0805 resistor	1	-
18.2k	0805 resistor	1	-
180k	0805 resistor	11	-
1M	0805 resistor	2	-
2.7M	0805 resistor	1	-
200k	0805 resistor	8	-
21k	0805 resistor	1	-
220k	0805 resistor	9	-
24.9k	0805 resistor	1	-
240k	0805 resistor	9	-
270k	0805 resistor	8	-
28k	0805 resistor	1	-
2M	0805 resistor	1	-

Value	Package	Qty	CSV tolerance
3.9M	0805 resistor	1	-
300k	0805 resistor	7	-
330k	0805 resistor	2	-
33k	0805 resistor	1	-
34k	0805 resistor	1	-
35.7k	0805 resistor	1	-
360k	0805 resistor	5	-
390k	0805 resistor	2	-
430k	0805 resistor	2	-
43k	0805 resistor	2	-
470k	0805 resistor	2	-
5.1M	0805 resistor	1	-
510k	0805 resistor	1	-
51k	0805 resistor	1	-
560k	0805 resistor	2	-
56k	0805 resistor	3	-
61.9k	0805 resistor	1	-
620k	0805 resistor	3	-
62k	0805 resistor	1	-
680k	0805 resistor	1	-
68k	0805 resistor	6	-
750k	0805 resistor	2	-
75k	0805 resistor	2	-
820k	0805 resistor	1	-
82k	0805 resistor	2	-
91k	0805 resistor	5	-
PinHeader_2x38_Straddle	2x38 straddle, 2.54 mm	1	-
TL074	SOIC-14, 1.27 mm	3	-

Every populated component

Ref	Value	Footprint / package	Side	Tol / status
C_U1N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U1P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U2N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U2P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U3N	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
C_U3P	0.1uF	Capacitor_SMD:C_0805_2012Metric	F	FIT
J_STRADDLE_EDGE	PinHeader_2x38_Straddle	Connector_EdgeMount:PinHeader_2x38_Straddle_P2.54mm	F	FIT
R002	91k	R_0805_2012Metric	F	FIT
R003	100k	R_0805_2012Metric	F	FIT
R006	1.6M	R_0805_2012Metric	F	FIT
R007	220k	R_0805_2012Metric	F	FIT
R010	270k	R_0805_2012Metric	F	FIT
R012	56k	R_0805_2012Metric	F	FIT
R014	56k	R_0805_2012Metric	F	FIT
R015	470k	R_0805_2012Metric	F	FIT
R018	180k	R_0805_2012Metric	F	FIT
R019	300k	R_0805_2012Metric	F	FIT
R021	240k	R_0805_2012Metric	F	FIT
R028	220k	R_0805_2012Metric	F	FIT
R032	82k	R_0805_2012Metric	F	FIT
R040	180k	R_0805_2012Metric	F	FIT
R041	110k	R_0805_2012Metric	F	FIT
R044	51k	R_0805_2012Metric	F	FIT
R046	300k	R_0805_2012Metric	F	FIT
R047	240k	R_0805_2012Metric	F	FIT
R050	240k	R_0805_2012Metric	F	FIT
R051	360k	R_0805_2012Metric	F	FIT
R054	430k	R_0805_2012Metric	F	FIT
R056	270k	R_0805_2012Metric	F	FIT
R057	5.1M	R_0805_2012Metric	F	FIT
R062	100k	R_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
R064	150k	R_0805_2012Metric	F	FIT
R065	91k	R_0805_2012Metric	F	FIT
R067	360k	R_0805_2012Metric	F	FIT
R070	130k	R_0805_2012Metric	F	FIT
R072	470k	R_0805_2012Metric	F	FIT
R074	68k	R_0805_2012Metric	F	FIT
R075	360k	R_0805_2012Metric	F	FIT
R078	200k	R_0805_2012Metric	F	FIT
R080	130k	R_0805_2012Metric	F	FIT
R082	1M	R_0805_2012Metric	F	FIT
R083	120k	R_0805_2012Metric	F	FIT
R085	680k	R_0805_2012Metric	F	FIT
R088	750k	R_0805_2012Metric	F	FIT
R090	120k	R_0805_2012Metric	F	FIT
R091	300k	R_0805_2012Metric	F	FIT
R094	300k	R_0805_2012Metric	F	FIT
R095	390k	R_0805_2012Metric	F	FIT
R098	62k	R_0805_2012Metric	F	FIT
R100	430k	R_0805_2012Metric	F	FIT
R102	56k	R_0805_2012Metric	F	FIT
R103	82k	R_0805_2012Metric	F	FIT
R106	130k	R_0805_2012Metric	F	FIT
R107	270k	R_0805_2012Metric	F	FIT
R109	220k	R_0805_2012Metric	F	FIT
R112	180k	R_0805_2012Metric	F	FIT
R114	68k	R_0805_2012Metric	F	FIT
R116	200k	R_0805_2012Metric	F	FIT
R119	180k	R_0805_2012Metric	F	FIT
R122	200k	R_0805_2012Metric	F	FIT
R124	43k	R_0805_2012Metric	F	FIT
R126	220k	R_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
R130	68k	R_0805_2012Metric	F	FIT
R131	100k	R_0805_2012Metric	F	FIT
R133	200k	R_0805_2012Metric	F	FIT
R136	330k	R_0805_2012Metric	F	FIT
R140	150k	R_0805_2012Metric	F	FIT
R142	75k	R_0805_2012Metric	F	FIT
R146	68k	R_0805_2012Metric	F	FIT
R147	270k	R_0805_2012Metric	F	FIT
R150	100k	R_0805_2012Metric	F	FIT
R151	68k	R_0805_2012Metric	F	FIT
R153	220k	R_0805_2012Metric	F	FIT
R156	150k	R_0805_2012Metric	F	FIT
R160	75k	R_0805_2012Metric	F	FIT
R161	240k	R_0805_2012Metric	F	FIT
R164	160k	R_0805_2012Metric	F	FIT
R166	200k	R_0805_2012Metric	F	FIT
R168	91k	R_0805_2012Metric	F	FIT
R170	620k	R_0805_2012Metric	F	FIT
R174	270k	R_0805_2012Metric	F	FIT
R175	330k	R_0805_2012Metric	F	FIT
R177	120k	R_0805_2012Metric	F	FIT
R179	300k	R_0805_2012Metric	F	FIT
R182	100k	R_0805_2012Metric	F	FIT
R183	91k	R_0805_2012Metric	F	FIT
R186	180k	R_0805_2012Metric	F	FIT
R188	270k	R_0805_2012Metric	F	FIT
R189	180k	R_0805_2012Metric	F	FIT
R191	1M	R_0805_2012Metric	F	FIT
R193	620k	R_0805_2012Metric	F	FIT
R195	620k	R_0805_2012Metric	F	FIT
R198	180k	R_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
R200	150k	R_0805_2012Metric	F	FIT
R204	68k	R_0805_2012Metric	F	FIT
R205	150k	R_0805_2012Metric	F	FIT
R207	240k	R_0805_2012Metric	F	FIT
R210	33k	R_0805_2012Metric	F	FIT
R212	220k	R_0805_2012Metric	F	FIT
R214	91k	R_0805_2012Metric	F	FIT
R215	180k	R_0805_2012Metric	F	FIT
R218	220k	R_0805_2012Metric	F	FIT
R220	110k	R_0805_2012Metric	F	FIT
R229	180k	R_0805_2012Metric	F	FIT
R232	240k	R_0805_2012Metric	F	FIT
R234	180k	R_0805_2012Metric	F	FIT
R242	510k	R_0805_2012Metric	F	FIT
R244	200k	R_0805_2012Metric	F	FIT
R245	1.1M	R_0805_2012Metric	F	FIT
R248	390k	R_0805_2012Metric	F	FIT
R249	200k	R_0805_2012Metric	F	FIT
R252	300k	R_0805_2012Metric	F	FIT
R253	180k	R_0805_2012Metric	F	FIT
R256	43k	R_0805_2012Metric	F	FIT
R260	160k	R_0805_2012Metric	F	FIT
R262	220k	R_0805_2012Metric	F	FIT
R263	820k	R_0805_2012Metric	F	FIT
R269	150k	R_0805_2012Metric	F	FIT
R272	270k	R_0805_2012Metric	F	FIT
R282	220k	R_0805_2012Metric	F	FIT
R284	100k	R_0805_2012Metric	F	FIT
R285	750k	R_0805_2012Metric	F	FIT
R289	130k	R_0805_2012Metric	F	FIT
R291	2.7M	R_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
R294	200k	R_0805_2012Metric	F	FIT
R295	240k	R_0805_2012Metric	F	FIT
R298	110k	R_0805_2012Metric	F	FIT
R299	300k	R_0805_2012Metric	F	FIT
R303	160k	R_0805_2012Metric	F	FIT
R312	240k	R_0805_2012Metric	F	FIT
R314	360k	R_0805_2012Metric	F	FIT
RB0P	270k	R_0805_2012Metric	F	FIT
RB1P	160k	R_0805_2012Metric	F	FIT
RB2P	560k	R_0805_2012Metric	F	FIT
RB3N	160k	R_0805_2012Metric	F	FIT
RB4N	560k	R_0805_2012Metric	F	FIT
RB5N	2M	R_0805_2012Metric	F	FIT
RB6P	240k	R_0805_2012Metric	F	FIT
RB7P	3.9M	R_0805_2012Metric	F	FIT
RB9N	360k	R_0805_2012Metric	F	FIT
RF0	100k	R_0805_2012Metric	F	FIT
RF1	100k	R_0805_2012Metric	F	FIT
RF2	100k	R_0805_2012Metric	F	FIT
RF3	100k	R_0805_2012Metric	F	FIT
RF4	100k	R_0805_2012Metric	F	FIT
RF5	100k	R_0805_2012Metric	F	FIT
RF6	100k	R_0805_2012Metric	F	FIT
RF7	100k	R_0805_2012Metric	F	FIT
RF8	100k	R_0805_2012Metric	F	FIT
RF9	100k	R_0805_2012Metric	F	FIT
RG0P	14.7k	R_0805_2012Metric	F	FIT
RG1P	24.9k	R_0805_2012Metric	F	FIT
RG2P	35.7k	R_0805_2012Metric	F	FIT
RG3P	61.9k	R_0805_2012Metric	F	FIT
RG4P	17.4k	R_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
RG5P	28k	R_0805_2012Metric	F	FIT
RG6P	14.3k	R_0805_2012Metric	F	FIT
RG7P	34k	R_0805_2012Metric	F	FIT
RG8P	21k	R_0805_2012Metric	F	FIT
RG9P	18.2k	R_0805_2012Metric	F	FIT
U1	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT
U2	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT
U3	TL074	Package_SO:SOIC-14_3.9x8.7mm_P1.27mm	F	FIT

DNP component sites - leave pads open

Every reference below has value DNP, footprint R_0805_2012Metric (0805 resistor), side F. They are excluded from grouped fitted quantities and the combined shopping totals.

R001, R004, R005, R008, R009, R011, R013, R016, R017, R020, R022, R023, R024, R025, R026, R027, R029, R030, R031, R033, R034, R035, R036, R037, R038, R039, R042, R043, R045, R048, R049, R052, R053, R055, R058, R059, R060, R061, R063, R066, R068, R069, R071, R073, R076, R077, R079, R081, R084, R086, R087, R089, R092, R093, R096, R097, R099, R101, R104, R105.

R108, R110, R111, R113, R115, R117, R118, R120, R121, R123, R125, R127, R128, R129, R132, R134, R135, R137, R138, R139, R141, R143, R144, R145, R148, R149, R152, R154, R155, R157, R158, R159, R162, R163, R165, R167, R169, R171, R172, R173, R176, R178, R180, R181, R184, R185, R187, R190, R192, R194, R196, R197, R199, R201, R202, R203, R206, R208, R209, R211.

R213, R216, R217, R219, R221, R222, R223, R224, R225, R226, R227, R228, R230, R231, R233, R235, R236, R237, R238, R239, R240, R241, R243, R246, R247, R250, R251, R254, R255, R257, R258, R259, R261, R264, R265, R266, R267, R268, R270, R271, R273, R274, R275, R276, R277, R278, R279, R280, R281, R283, R286, R287, R288, R290, R292, R293, R296, R297, R300, R301.

R302, R304, R305, R306, R307, R308, R309, R310, R311, R313, R315, R316, R317, R318, R319, R320, RB0N, RB1N, RB2N, RB3P, RB4P, RB5P, RB6N, RB7N, RB8N, RB8P, RB9P, RG0N, RG1N, RG2N, RG3N, RG4N, RG5N, RG6N, RG7N, RG8N, RG9N.

motherboard_v2

2 copper layers | 100 x 100 mm | 81 fitted items | 0 DNP sites

MANDATORY BOTTOM BODGE WIRE: JS1 pad B7 to JS2 pad B7. Fit before powering the stack.

72 top-mounted items, 9 back-mounted items. THT connectors on the top are soldered at their pins on the back. The two placement diagram pages follow this section.

Verified source: /home/kit/mb-jd0-mask/post_crb.kicad_pcb
SHA-256 4932b44c056e39017216b71dfa885c8117e32a04a94834f4cd051757efb9d1ba
Provenance note: fab quote-details.json retained the pre-mask-fix hash 271faeaa... . The authoritative fixed source hash above matches mb-jd0-mask/verification.json. No older source was used.
Mounting holes (excluded from parts totals): H1 at (5, 5) mm; H2 at (95, 5) mm; H4 at (95, 95) mm; H3 at (5, 95) mm.

Grouped fitting summary

Value	Package	Qty	CSV tolerance
0.1uF	0805 capacitor	3	-
10uF	0805 capacitor	1	-
1uF	0805 capacitor	1	-
0R	0805 resistor	1	-
1k	0805 resistor	10	-
220k	0805 resistor	10	-
22k	0805 resistor	20	-
820	0805 resistor	1	-
2x38_2.54mm_GENERIC_MANUAL_SOCKET	2x38 socket, 2.54 mm	1	-
Slot0: DC1A (L1 Neurons 0..3)	2x38 socket, 2.54 mm	1	-
Slot1: DC1B (L1 Neurons 4..7)	2x38 socket, 2.54 mm	1	-
Slot2: DC1C (L1 Neurons 8..11)	2x38 socket, 2.54 mm	1	-
Slot3: DC1D (L1 Neurons 12..15)	2x38 socket, 2.54 mm	1	-
Slot4: DC2 (16x Active Precision ReLU)	2x38 socket, 2.54 mm	1	-
Slot5: DC3 (L2 Weights + 10-Way WTA)	2x38 socket, 2.54 mm	1	-
100uF	CP_Elec_8x10	2	-
3V3_GND_FROM_MCU	PinHeader_1x02_P2.54mm_Vertical	1	-
+15V_GND_-15V	PinHeader_1x03_P2.54mm_Vertical	1	-
MCU_3V3_SPI_PIN1_NC	PinHeader_1x06_P2.54mm_Vertical	1	-
Scores_DIV_3V3_Out	PinHeader_1x12_P2.54mm_Vertical	1	-
1N4148W	SOD-123	20	-
L7805CV	TO-220-3_Vertical	1	-

TOP side

Ref	Value	Footprint / package	Side	Tol / status
DH0	1N4148W	D_SOD-123	F	FIT
DH1	1N4148W	D_SOD-123	F	FIT
DH2	1N4148W	D_SOD-123	F	FIT
DH3	1N4148W	D_SOD-123	F	FIT
DH4	1N4148W	D_SOD-123	F	FIT
DH5	1N4148W	D_SOD-123	F	FIT
DH6	1N4148W	D_SOD-123	F	FIT
DH7	1N4148W	D_SOD-123	F	FIT
DH8	1N4148W	D_SOD-123	F	FIT
DH9	1N4148W	D_SOD-123	F	FIT
DL0	1N4148W	D_SOD-123	F	FIT
DL1	1N4148W	D_SOD-123	F	FIT
DL2	1N4148W	D_SOD-123	F	FIT
DL3	1N4148W	D_SOD-123	F	FIT
DL4	1N4148W	D_SOD-123	F	FIT
DL5	1N4148W	D_SOD-123	F	FIT
DL6	1N4148W	D_SOD-123	F	FIT
DL7	1N4148W	D_SOD-123	F	FIT
DL8	1N4148W	D_SOD-123	F	FIT
DL9	1N4148W	D_SOD-123	F	FIT
JAD	Scores_DIV_3V3_Out	Connector_PinHeader_2.54mm:PinHeader_1x12_P2.54mm_Vertical	F	FIT
JAR	3V3_GND_FROM_MCU	PinHeader_1x02_P2.54mm_Vertical	F	FIT
JD0	2x38_2.54mm_GENERIC_MANUAL_SOCKET	DC0_Isolated:Socket2x38	F	FIT
JP	+15V_GND_-15V	PinHeader_1x03_P2.54mm_Vertical	F	FIT
JS0	Slot0: DC1A (L1 Neurons 0..3)	Connector_PinSocket_2.54mm:PinSocket_2x38_P2.54mm_Vertical	F	FIT
JS1	Slot1: DC1B (L1 Neurons 4..7)	Connector_PinSocket_2.54mm:PinSocket_2x38_P2.54mm_Vertical	F	FIT
JS2	Slot2: DC1C (L1 Neurons 8..11)	Connector_PinSocket_2.54mm:PinSocket_2x38_P2.54mm_Vertical	F	FIT
JS3	Slot3: DC1D (L1 Neurons 12..15)	Connector_PinSocket_2.54mm:PinSocket_2x38_P2.54mm_Vertical	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
JS4	Slot4: DC2 (16x Active Precision ReLU)	Connector_PinSocket_2.54mm:PinSocket_2x38_P2.54mm_Vertical	F	FIT
JS5	Slot5: DC3 (L2 Weights + 10-Way WTA)	Connector_PinSocket_2.54mm:PinSocket_2x38_P2.54mm_Vertical	F	FIT
JSPI	MCU_3V3_SPI_PIN1_NC	Connector_PinHeader_2.54mm:PinHeader_1x06_P2.54mm_Vertical	F	FIT
RA0	1k	Resistor_SMD:R_0805_2012Metric	F	FIT
RA1	1k	Resistor_SMD:R_0805_2012Metric	F	FIT
RA2	1k	Resistor_SMD:R_0805_2012Metric	F	FIT
RA3	1k	Resistor_SMD:R_0805_2012Metric	F	FIT
RA4	1k	Resistor_SMD:R_0805_2012Metric	F	FIT
RA5	1k	Resistor_SMD:R_0805_2012Metric	F	FIT
RA6	1k	Resistor_SMD:R_0805_2012Metric	F	FIT
RA7	1k	Resistor_SMD:R_0805_2012Metric	F	FIT
RA8	1k	Resistor_SMD:R_0805_2012Metric	F	FIT
RA9	1k	Resistor_SMD:R_0805_2012Metric	F	FIT
RB0	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RB1	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RB2	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RB3	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RB4	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RB5	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RB6	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RB7	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RB8	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RB9	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RR0	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RR1	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RR2	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RR3	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RR4	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RR5	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RR6	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RR7	22k	Resistor_SMD:R_0805_2012Metric	F	FIT

Ref	Value	Footprint / package	Side	Tol / status
RR8	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RR9	22k	Resistor_SMD:R_0805_2012Metric	F	FIT
RT0	220k	Resistor_SMD:R_0805_2012Metric	F	FIT
RT1	220k	Resistor_SMD:R_0805_2012Metric	F	FIT
RT2	220k	Resistor_SMD:R_0805_2012Metric	F	FIT
RT3	220k	Resistor_SMD:R_0805_2012Metric	F	FIT
RT4	220k	Resistor_SMD:R_0805_2012Metric	F	FIT
RT5	220k	Resistor_SMD:R_0805_2012Metric	F	FIT
RT6	220k	Resistor_SMD:R_0805_2012Metric	F	FIT
RT7	220k	Resistor_SMD:R_0805_2012Metric	F	FIT
RT8	220k	Resistor_SMD:R_0805_2012Metric	F	FIT
RT9	220k	Resistor_SMD:R_0805_2012Metric	F	FIT
UR5	L7805CV	TO-220-3_Vertical	F	FIT

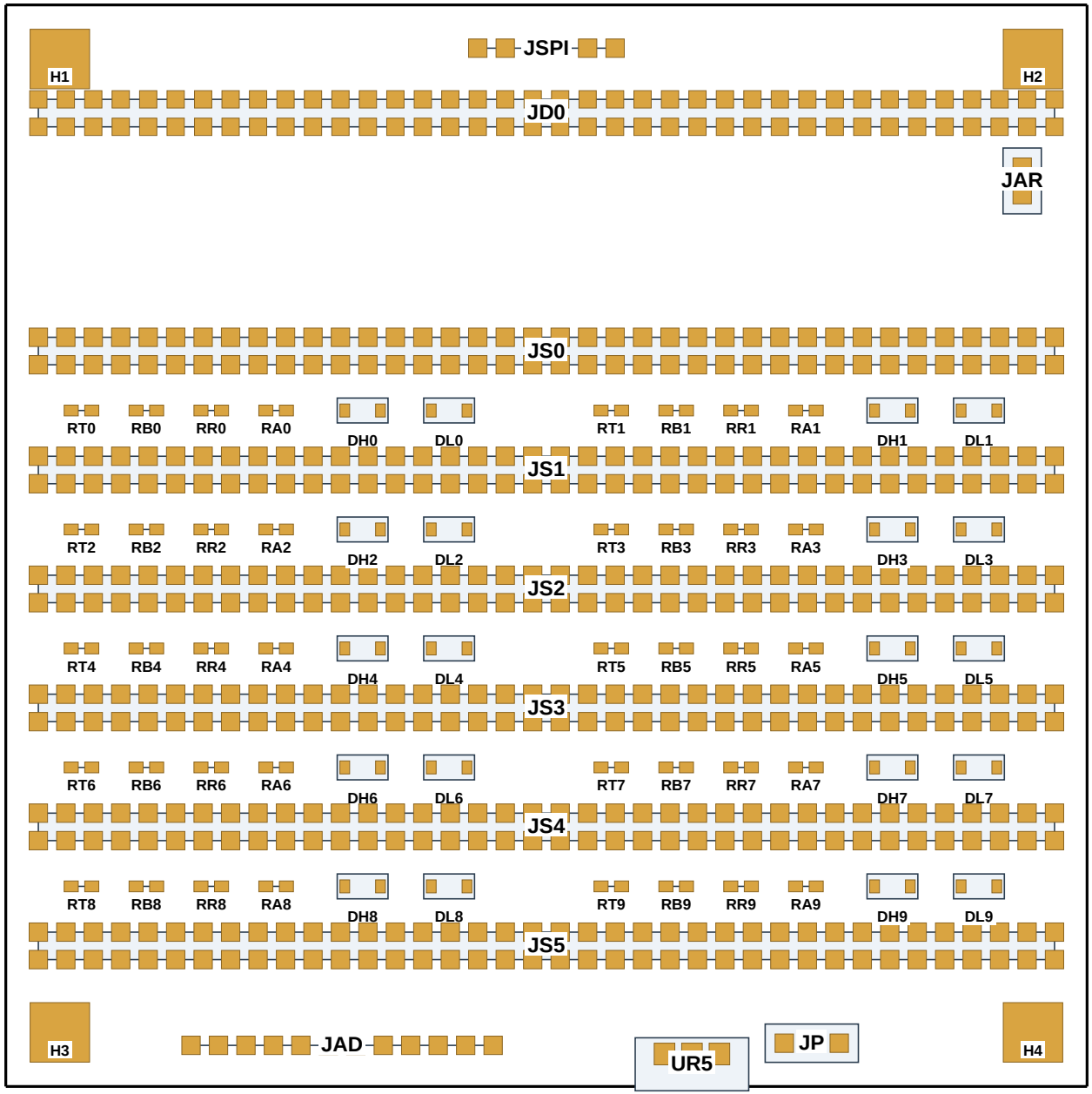
BACK side

Ref	Value	Footprint / package	Side	Tol / status
CBN	100uF	CP_Elec_8x10	B	FIT
CBP	100uF	CP_Elec_8x10	B	FIT
CRB	10uF	Capacitor_SMD:C_0805_2012Metric	B	FIT
CRI	1uF	C_0805_2012Metric	B	FIT
CRN	0.1uF	C_0805_2012Metric	B	FIT
CRO	0.1uF	C_0805_2012Metric	B	FIT
CRP	0.1uF	C_0805_2012Metric	B	FIT
RM	820	R_0805_2012Metric	B	FIT
RRL	0R	R_0805_2012Metric	B	FIT

pcbai motherboard · TOP side (F.Cu), viewed from above

source post_crb.kicad_pcb sha256 4932b44c…d1ba · 76 parts on this side · board 100.0 × 100.0 mm · 1 mm = 4.66 pt

Reminder: mandatory JS1 B7 → JS2 B7 bodge wire goes on the BOTTOM (see page 2).

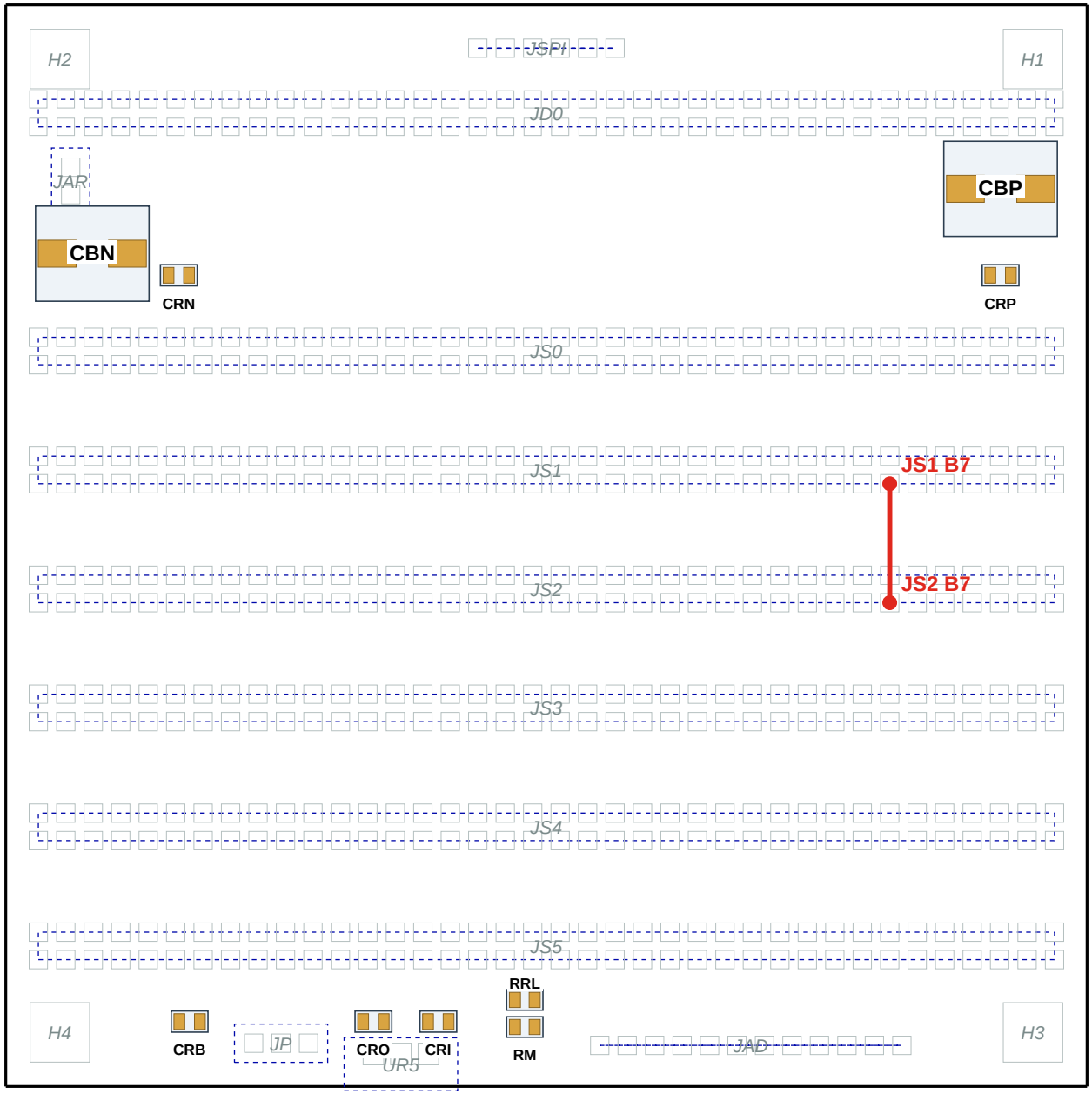


pads (gold) and outlines from KiCad footprint pads/CrtYd/Fab geometry; no coordinates

pcbai motherboard · BOTTOM side (B.Cu), MIRRORED: viewed from below

source post_crb.kicad_pcb sha256 4932b44c…d1ba · 9 parts on this side · board 100.0 × 100.0 mm · 1 mm = 4.66 pt

MANDATORY BOTTOM BODGE: wire JS1 pad B7 to JS2 pad B7 (red). Grey dashed = top-mounted THT parts, pins seen through.



Bodge endpoints (board mm, KiCad coords): JS1 B7 at (18.25, 44.27), JS2 B7 at (18.25, 55.27). Straight line shown is schematic; route the wire neatly.

pads (gold) and outlines from KiCad footprint pads/CrtYd/Fab geometry; no coordinates

Combined fitting totals - one complete stack

DC0 x1, DC1a x1, DC1b x1, DC1c x1, DC1d x1, DC2 x1, DC3 x1, motherboard x1. Excludes DNP, holes, spare bare PCBs and optional heatsinks. Add one insulated bottom bodge wire for JS1 B7 to JS2 B7. Values remain source values, not the purchased-kit rounded BOM.

Value	Package	Qty	Value	Package	Qty	Value	Package	Qty
0.1uF	0805 capacitor	41	22.6k	0805 resistor	1	6.2M	0805 resistor	1
10uF	0805 capacitor	1	220k	0805 resistor	46	61.9k	0805 resistor	2
1uF	0805 capacitor	65	22k	0805 resistor	20	620k	0805 resistor	11
0R	0805 resistor	1	24.9k	0805 resistor	1	62k	0805 resistor	10
1.1M	0805 resistor	12	240k	0805 resistor	34	680k	0805 resistor	16
1.2M	0805 resistor	6	24k	0805 resistor	1	68k	0805 resistor	14
1.3M	0805 resistor	7	270k	0805 resistor	46	750k	0805 resistor	18
1.5M	0805 resistor	4	27k	0805 resistor	3	75k	0805 resistor	19
1.6M	0805 resistor	4	28k	0805 resistor	1	8.66k	0805 resistor	1
1.8M	0805 resistor	3	29.4k	0805 resistor	1	820	0805 resistor	1
100k	0805 resistor	45	2M	0805 resistor	8	820k	0805 resistor	11
10k	0805 resistor	97	3.3M	0805 resistor	2	82k	0805 resistor	17
110k	0805 resistor	29	3.9M	0805 resistor	1	90.9k	0805 resistor	1
113k	0805 resistor	1	300k	0805 resistor	35	910k	0805 resistor	9
120k	0805 resistor	17	30k	0805 resistor	1	91k	0805 resistor	25
130k	0805 resistor	25	330k	0805 resistor	41	2x38_2.54mm_GENERIC_MANUAL_SOCKET	2x38 socket, 2.54 mm	1
13k	0805 resistor	1	33k	0805 resistor	4	Slot0: DC1A (L1 Neurons 0..3)	2x38 socket, 2.54 mm	1
14.3k	0805 resistor	2	34k	0805 resistor	1	Slot1: DC1B (L1 Neurons 4..7)	2x38 socket, 2.54 mm	1
14.7k	0805 resistor	2	35.7k	0805 resistor	2	Slot2: DC1C (L1 Neurons 8..11)	2x38 socket, 2.54 mm	1
15.4k	0805 resistor	1	360k	0805 resistor	28	Slot3: DC1D (L1 Neurons 12..15)	2x38 socket, 2.54 mm	1
150k	0805 resistor	34	36k	0805 resistor	3	Slot4: DC2 (16x Active Precision ReLU)	2x38 socket, 2.54 mm	1
16.2k	0805 resistor	1	390k	0805 resistor	24	Slot5: DC3 (L2 Weights + 10-Way WTA)	2x38 socket, 2.54 mm	1
160k	0805 resistor	38	39k	0805 resistor	6	2x38_2.54mm_GENERIC_MANUAL_STRADDLE	2x38 straddle, 2.54 mm	1
17.4k	0805 resistor	1	3M	0805 resistor	1	PinHeader_2x38_Straddle	2x38 straddle, 2.54 mm	6
18.2k	0805 resistor	3	430k	0805 resistor	24	100uF	CP_Elec_8x10	2
180k	0805 resistor	38	43k	0805 resistor	7	3V3_GND_FROM_MCU	PinHeader_1x02_P2.54mm_Vertical	1
1M	0805 resistor	10	470k	0805 resistor	25	+15V_GND_-15V	PinHeader_1x03_P2.54mm_Vertical	1
1k	0805 resistor	10	47k	0805 resistor	8	MCU_3V3_SPI_PIN1_NC	PinHeader_1x06_P2.54mm_Vertical	1
2.2M	0805 resistor	4	5.1M	0805 resistor	1	Scores_DIV_3V3_Out	PinHeader_1x12_P2.54mm_Vertical	1
2.4M	0805 resistor	1	510k	0805 resistor	20	1N4148W	SOD-123	52
2.7M	0805 resistor	1	51k	0805 resistor	8	TL074	SOIC-14, 1.27 mm	15
200k	0805 resistor	43	52.3k	0805 resistor	1	74AHCT595S16-13	SOIC-16 / SOP-16, 1.27 mm	8
20k	0805 resistor	1	560k	0805 resistor	21	L7805CV	TO-220-3_Vertical	1
21k	0805 resistor	1	56k	0805 resistor	9			

Total fitted items: 1237. Includes 15 TL074s, 8 shift registers, 52 1N4148W diodes and 65 1uF capacitors. Connector functional labels remain separate source-value rows.